Supplementary Table 1: Characteristics of studies included in this meta-analysis

Author, years, country	Design	Study description	Study periods	Study size, no. of death	Exposure	Comparison	RR (95 % CI)	Adjustment factors
Abohamr et al 2020, Saudi Arabia	Retrospective cohort study	King Saud Medical City, Riyadh	March 23 and June 15, 2020	768 hospitalized patients with COVID-19; 89 died	Diabetes Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no	3.01 (1.867-4.84) 18.143 (10.862-30.30) 7.018 (4.398-11.199)	None
Adrish et al, 2020, USA	Retrospective cohort study	Data from the Bronx Care Health System, the largest voluntary, not-for- profit health and teaching hospital system, serving the south and central Bronx in the New York City	March 9 to May 18, 2020	1173 COVID-19 patients; 390 died	Smoking	Ever smokers vs. never Current smokers vs. never Past smokers vs. never	1.34 (1.08-1.66) 1.62 (1.22-2.13) 1.15 (0.87-1.51)	Age
Alamdari et al, 2020, Iran	Retrospective cohort study	Patients admitted Shahid Modarres Hospital in Tehran,	April 2 and May 3	157 COVID-19 patients; 18 died	BMI	>30 kg/m2	2.53 (1.08-5.92)	None
Albitar et al, 2020, China	cross-sectional study	Data from national, provincial, and municipal health reported from Hubei and outside of Hubei	December 1, 2019 to April 21, 2020	828 COVID-19 patients; 219 died	Diabetes Hypertension	Yes vs. no Yes vs. no	12.23 (4.13-36.27) 3.58 (1.69-7.55)	Age, sex, geographic location and chronic disease (hypertension, diabetes, chronic lung diseases, chronic kidney and cardiovascular disease)
Alharthy et al, 2020, Saudi Arabia	Retrospective cohort study	Patients from King Saud Medical City (KSMC), Riyadh (adult > 18 years old, and admitted to ICU)	March 20 and May 31, 2020.	352 COVID-19 patients; 114 died	Smoking	Active smokers vs. never	3.00 (2.51–3.66)	Age, ICU length of stay, Spo_2/FiO_2 ratio, white blood cells, lymphocytes, d-dimers, lactate and pulmonary embolism
Almazeedi et al 2020, Kuwait	Retrospective cohort study	Data from JaberAl-Ahmad Al-Sabah hospitalisa (1240 bed tertiary hospital based in South Surra)	February 24 and April 20, 2020	1096 hospitalized patients with COVID-19; 19 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	0.83 (0.17-4.16) 0.84 (0.46-4.81) 0.22 (0.03-1.51) 10.09 (1.22-83.4)	age, sex, hypertension, smoking, diabetes, asthma, chronic renal disease, qSOFA score, elevated CRP and elevated procalcitonin
Al-Salameh et al, 2020, France	Observational cohort study	Amiens University Hospital	case confirmed until April 21 2020	433 COVID-19 patients; 21 died	Diabetes	Yes vs. no	0.73 (0.40-1.34)	sex, age, the presence/absence of hypertension, the presence/absence of

								cardiovascular or cardiac disease, the presence/absence of obesity, the presence/absence of chronic obstructive pulmonary disease (COPD) or sleep apnoea syndrome, a history of chronic kidney disease (CKD), a history of cancer, treatment with ACEIs or ARBs, glycaemia on admission, the white blood cell (WBC) count on admission, the glomerular filtration rate on admission, the serum C-reactive protein (CRP) on admission, and abnormal liver function test results
An et al 2020, South Korea	Retrospective cohort study	Data provided by the Korean National Health Insurance Service (KNHIS): the database of beneficiaries of national health insurance	January 23 andApril 2, 2020	10,237 hospitalized patients with COVID-19; 228 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.75 (1.29-2.36) 1.22 (0.87-1.73)	Age group, sex, income level, residence, household type, disablility, symptom and infection route.
Anderson et al, 2020, USA	Retrospective cohort study	Patients admitted from the emergency department (ED) to NewYork-Presbyterian/Columbia University Irving Medical Center	March 10 and 24 April 2020	2466 COVID-19 patients; 627 died	BMI	<18.5 kg/m2 18.5-24.9 kg/m2 25-29.9 kg/m2 30-34.9 kg/m2 35-39.9 kg/m2 ≥40 kg/m2	2.4 (1.1–5.3) 1.5 (1.02–2.2) 1 (reference) 1.4 (0.97–2.0) 1.0 (0.6–1.7) 1.6 (0.9–2.7)	Age, sex, race/ethnicity, hypertension, asthma or chronic obstructive pulmonary disease, chronic kidney disease, pulmonary hypertension, smoking, cancer, and diabetes
Anzola et al, 2020, Italy	Observational cohort study	Data from the Hospital of Gavardo, located in the neighborhood of Brescia in	March 12, 2020 to April 11, 2020	431 COVID-19 patients; 72 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.13 (0.46-2.73) 0.60 (0.27-1.34)	Age, creatinine and CRP

		Lombardia						
Arshad et al 2020, USA	Retrospective cohort study	Data from the Henry Ford Health System (HFHS) in Southeast Michigan, a large six hospital integrated health system; the largest of hospital sisan802-bed quaternary academic teaching hospital in urban Detroit, Michigan.	March 10 and May 2, 2020	2,541 hospitalized patients with COVID-19; 460 died	Diabetes Hypertension Obesity	Yes vs no Yes vs no Yes vs no	0.98 (0.79-1.211) 0.80 (0.63-1.01) 0.78 (0.62-0.96)	Age, sex, white race, lung disease, BMI, COPD, hypertension, cancer, diabetes, admitted to ICU, ventilator and admission factors
Atkins et al, 2020, UK	Observational cohort study	UK Biobank Community Cohort; aged 65–86 years	March 16 to April 26, 2020.	507 COVID-19 patients; 41 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.11 (2.06-4.71) 1.42 (0.96-2.11)	Age group, sex, ethnicity, education, baseline assessment center, hypertension, diabetes and others chronic diseases
Barbu et al, 2020, Romania	Retrospective cohort study	Data from ministry of Health; 61.4% of males and 38.6% females; 68.2 years	March 22 to May 8, 2020	814 died	Diabetes Hypertension	Yes vs. no Yes vs. no	6.43 (4.97-8.32) 3.26 (2.69-3.96)	Age
Baronio et al 2020, Italy	Retrospective cohort study	Patients admitted to the “Fondazione Poliambulanza di Brescia” hospital (Brescia, Lombardy, Italy)	March 1, and April 11, 2020	191 hospitalized patients with COVID-19; 78 died	Obesity	18.5-<25 25-<30 30+	1 1.36 (0.70-2.65) 2.23 (1.15-4.35)	Age, sex, cardiovascular plus hypertension and temperature
Barron et al, 2020, England	Retrospective cohort study	Data from whole-population study (NHS England and NHS Improvement created a bespoke hub of relevant datasets in the National Commissioning Data Repository) covering all individuals registered with a general practice in England; 40.9 years	March 1 to May 11 2020	23 698 in-hospital COVID-19-related deaths	Diabetes	Yes vs. no	2.03 (1.97-2.09)	Age, sex, ethnicity, deprivation quintile and region.
Bepouka et al 2020,	Retrospective cohort study	Data from Kinshasa University Hospital; the tertiary level	March 23 and June 15, 2020	141 hospitalized patients with COVID-19; 41 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.31 (0.59-2.88) 1.07 (0.51-2.26)	Age, sex, civil status, admission procedure,

Democratic Republic of the Congo		hospital in the city of Kinshasa with a capacity of 1000 beds						comorbidities, all admission signs, blood oxygen saturation, severity of illness, conventional treatment and biologic variables
Berenguer et al 2020, Spain	Retrospective observational study	Data from the COVID-19-Spain, a study of 127 Spanish centres until 17 March 2020		4035 hospitalized patients with COVID-19; 1131 died	Diabetes Hypertension Obesity	Yes vs no Yes vs no Yes vs no	1.23 (1.07-1.41) 1.81 (1.56-2.09) 1.21 (1.01-1.44)	Age, sex, hypertension, diabetes, obesity, cancer, others chronic diseases, admission signs and symptoms and vital signs.
Biscarini et al 2020, Italy	Retrospective cohort study	Data fro the SMAtteo Covid 19 Registry (SMACORE) cohort (Hospital, Fondazione IRCCS Policlinico San Matteo in Pavia)	February 21 and March 31, 2020	427 hospitalized patients with COVID-19; 139 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.29 (0.78-2.15) 0.77 (0.51-1.17) 1.03 (0.65-1.67)	Sex, age, presence of interstitial pneumonia at admission, CRP levels and PF ratio at admission, antiviral and antibiotic treatment, and main comorbidities (tumor, heart disease, hypertension, diabetes)
Boulle et al, 2020, South Africa	Cohort study	Data from the Western Cape Provincial Health Data Centre (WCPHDC) of public sector patients; aged ≥20	before March 1, 2020	15,203 COVID-19 patients; 625 died	Diabetes Hypertension Obesity	diabetes HbA1c <7% vs. no diabetes HbA1c 7 - 8.9% vs. no diabetes HbA1c ≥9% vs. no diabetes no HbA1c measurement vs. no Yes vs. no	2.21 (1.57-3.12) 3.41 (2.59-4.51) 3.62 (2.8- 4.59) 2.02 (1.47-2.76) 1.02 (0.84-1.24)	Age, sex, diabetes, hypertension, HIV, chronic kidney disease, chronic pulmonary diseases and tuberculosis.
Bousquet et al, 2020	Retrospective cohort study	The university hospital of Paris-Seine Saint Denis	March to April 2020	108 COVID-19 patients; 28 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.57 (0.72-3.44) 1.34 (0.54-3.32)	Age

France		(Avicenne hospital; nearly 200 hospital beds for Covid-19 patients); aged 65			BMI	BMI < 21 kg/m2	0.89 (0.34-2.35)	
Brouns et al 2020, Netherlands	Retrospective cohort study	Data from 14 NH facilities from the NH organization Envida, Maastricht,	March 20 and May 1, 2020	67 hospitalized patients with COVID-19; 35 died	Hypertension	Yes vs. no	0.54 (0.22-1.37)	Age, sex, COVID-19 rRT-PCR result, CCI, hypertension, medication use, and OAT
Calmes et al 2020, Belgium	Retrospective cohort study	Patients hospitalized in University Hospital of Liege	March 18 and April 17, 2020	595 hospitalized patients with COVID-19; 17 died	Diabetes Hypertension Smoking BMI Obesity	Yes vs. no Yes vs. no smoker or ex-smoker vs never BMI per Yes vs. no	1.80 (1.10-3.10) 1.40 (0.80-2.30) 1.40 (0.81-2.30) 1.00 (0.97-1.10) 1.80 (1.10-3.20)	Age and gender
Cao et al, 2020 China	Retrospective cohort study	Wuhan Jin Yin-tan hospital and Wuhan Union hospital	January 5, and February 22, 2020	101 hospitalized patients with COVID-19; 66 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.24 (0.23-6.64) 0.73 (0.15-3.56)	Age, respiratory rate, acute respiratory distress syndrome, cardiovascular disease, hypertension, diabetes, chronic pulmonary disease and bacterial infection
Carrasco-Sanchez et al, 2020 Spain	Retrospective multi-center study	Data from the Spanish SEMI-COVID-19 Registry	March 1 to May 31, 2020	11312 hospitalized patients with COVID-19; 2289 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.63 (1.49-1.79) 1.14 (1.01-1.29)	Age, gender, hypertension, diabetes, COPD, dependency, lymphopenia, anemia (hemoglobin<10 g/dl), serum creatinine, CRP>60 mg/L, LDH>400 U/L, and Ddimer> 1000 ng/ml, elevated admission BG levels
Castelnuovo et al 2020, Italy	Retrospective cohort study	CORIST Collaboration Project (median age 67 years)	February 19 to May 23 2021	3894 hospitalized patients with COVID-19; 712 died	Diabetes Hypertension Smoking Obesity	Yes vs. no Yes vs. no Current smoker vs never ≥30 vs <30 kg/m2	1.03 (0.84-1.28) 0.87 (0.72-1.05) 0.94 (0.69-1.27) 1.21 (0.91-1.61)	Gender, age, diabetes, hypertension, history of myocardial infarction, heart failure, chronic pulmonary disease, chronic kidney disease, CRP, obesity, smoking habit and anti-COVID19

								drugs during hospitalization as fixed effects and repeated measures
Chachkhiani et al 2020 USA	Retrospective cohort study	University Medical Center New Orleans (UMCNO); aged (60 mean)	March 1, through March 31	250 hospitalized patients with COVID-19; 56 died	Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no Smoker vs. never per 1 kg/m2 increase	1.10 (0.60-1.80) 1.70 (0.80-3.70) 0.50 (0.10-1.90) 1.001 (1.000-1.002)	None
Chang et al 2020 South Korea	Retrospective cohort study	Kyungpook National University Chilgok Hospital and Kyungpook National University Hospital in Daegu; aged 67 (mean)	February 1 to April 10, 2020	106 hospitalized patients with COVID-19; 30 died	Diabetes	Yes vs. no	1.16 (0.51-2.65)	Age, chronic lung disease, fasting plasma glucose and white blood cell
Chen et al 2020 China	Retrospective cohort study	Data from Union Hospital (main part, Union West Hospital and Union Tumor Hospital), Wuhan Central Hospital, General Hospital of Central Theater Command, Wuhan Third Hospital and Wuhan Jin-Yin-Tan Hospital; ≥18 years patients	January 20 to April 4, 2020	1859 hospitalized patients with COVID-19; 208 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Smokers (current or former smoker) vs. never	2.47 (1.82-3.34) 2.21 (1.68–2.90) 1.84 (1.17–2.92)	Age
Chen et al 2020 China	Retrospective cohort study	Admitted to the Central Hospital of Wuhan (age : 15–99 years)	1 January 2020 to March, 3 2020	904 hospitalized patients with COVID-19; 92 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.51 (1.53-4.13) 3.48 (2.24-5.40)	None
Chen et al 2020, China	Retrospective cohort study	Data from patients admitted to The Central Hospital ofWuhan	1 January 2020 to 15 February 2020	660 hospitalized patients with COVID-19; 82 died	Hypertension	Yes vs. no	0.92 (0.42-2.02)	Age, cerebral infraction, SOFA, CRP and LDH
Chilimuri et al 2020 USA	Retrospective cohort study	The Bronxcare Health System (BCHS) hospital in the South Bronx, New York	March 9 to April 9, 2020	375 hospitalized patients with COVID-19; 160 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.94-2.65) 1.46 (0.82-2.62)	several variables (age, sex, hypertension, diabetes, CVD, chronic kidney disease and biological measurement)
Ciardullo et al 2020 Italy	Retrospective cohort study	Policlinico di Monza hospital in Monza (aged ≥ 18 years)	February 22 and May 15, 2020	373 hospitalized patients with COVID-19; 142 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.23 (0.85-1.63) 1.32 (0.91-1.81)	Age, sex, chronic kidney disease, hypertension diabetes, COPD, chronic obstructive and

								pulmonary disease
Ciceri et al 2020, Spain	Retrospective cohort study	Data from North-East of Milan's province and admitted at San Raffaele Hospital; aged ≥18 years	February 25th and March 24th	410 hospitalized patients with COVID-19; 95 died	Diabetes Hypertension BMI (not precised)	Yes vs no Yes vs no	1.51 (0.96–2.05) 2.60 (1.67–4.05) 0.96 (0.90–1.02)	Age
Coppelli et al 2020 Italy	Retrospective cohort study	The Pisa COVID-19 Study (University Uospital, Pisa)	20 March to 30 April 2020	271 hospitalized patients with COVID-19; 67 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no BMI >30 kg/m2 Smokers vs non-smokers	0.78 (0.29-2.09) 1.02 (0.49-2.15) 2.80 (0.97-8.10) 0.74 (0.37-1.47)	Age, sex, hypertension, cerebrovascular disease, chronic obstructive pulmonary disease, chronic kidney disease, cognitive impairment and several biomarkers
Crouse et al 2020 USA	Retrospective cohort study	University of Alabama at Birmingham Hospital (a tertiary health care center in the racially diverse Southern U.S.)	25 February to June 22, 2020	604 hospitalized patients with COVID-19; 45 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	3.62 (2.11-6.20) 3.60 (1.68-7.69) 1.14 (0.67-1.94)	Age, race, sex, obese status and hypertension status
Cummings et al 2020 USA	Prospective cohort study	Data from two NewYork-Presbyterian hospitals affiliated with Columbia University Irving Medical Center in northern Manhattan (aged ≥18 years)	March 2 to April 1, 2020	257 hospitalized patients with COVID-19; 101 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.31 (0.81-2.10) 1.58 (0.89-2.81) 0.76 (0.40-1.47)	Nine independent variables (age, sex, hypertension, COPD, chronic kidney disease, diabetes, symptom duration before hospital, chronic cardiac disease and BMI)
Czernichow et al 2020, France	Prospective cohort study	Data from the Paris hospitals, France	February 1 and 30 April 2020	5795 hospitalized patients with COVID-19; 891 died	Diabetes Hypertension Smoking Obesity	Yes vs. no Yes vs. no Yes vs. no <18.5 18.5-24.9 25-29.99 30-34.99 35-39.99 40+	1.34 (1.11-1.62) 1.01 (0.82-1.24) 1.06 (0.84-1.33) 1.24 (0.77-1.93) 1 1.41 (1.21-1.77) 1.89 (1.45-2.47) 2.79 (1.95-3.97) 2.55 (1.62-3.95)	Comorbidities, smoking status, age and sex using logistic regressions, both including and excluding patients with missing BMI and with stratification on age class
De Souza et al 2020 Brazil	Cross-sectional study	Data from the state of Alagoas in the northeastern of Brazil (older adults (aged ≥60 years)	until 01 August, 2020	9807 hospitalized patients with COVID-19; 1171 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	2.33 (1.99-2.74) 1.53 (1.20-1.94) 1.77 (0.84–3.74)	Age, sex, initial symptoms reporte and comorbidities
De Vito et al	Retrospective	Patients admitted to the Unit of	March 8 to April	87 hospitalized patients	Hypertension	Yes vs. no	0.47 (0.11-2.00)	Age, moderate ARDS,

2020 Italy	cohort study	Infectious Diseases of the University Hospital of Sassari (North-West Sardinia, Italy)	8, 2020	with COVID-19; 57 died				not compliance patients lymphocytes and others comorbidities
Deng et al 2020, China	Retrospective cohort study	Data reported publicly by the Chinese center for disease control and prevention	updated through February 11, 2020	44,672 hospitalized patients with COVID-19; 1023 died	Diabetes Hypertension	Yes vs no Yes vs no	4.47 (3.49–5.61) 4.48 (3.69–5.45)	None
Dennis et al 2020 England	Retrospective cohort study	Patients admitted to hospital in England with COVID-19 requiring admission to a high dependency unit (HDU) or intensive care unit (ICU); aged 18–99	1 March 2020 and 27 July 2020	19256 hospitalized patients with COVID-19; 5077 died	Diabetes	Yes vs. no	1.23 (1.14-1.32)	Age, sex, ethnicity, obesity and comorbidity including hypertension and others
Desai et al 2020 Italy	Retrospective cohort study	Data from emergency Department (ED) of Humanitas Clinical and Research Center; aged 64.8 years (mean)	February 21 and April 14, 2020	372 hospitalized patients with COVID-19; 120 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.50-2.40) 0.90 (0.50-1.70)	Age, gender, chronic use of ACE-I and ARBs), time interval between onset of symptoms and ED admission.
Docherty et al 2020, UK	Prosepctive cohort study	Data from the ISARIC WHO CCP-UK study which is an ongoing study in 208 acute care hospitals in England, Scotland, and Wales	February and 19 April 2020	15 194 hospitalized patients with COVID-19; 3911 died	Diabetes Obesity Smoking	Yes vs. no Yes vs. no current vs. never Former vs. never	1.06 (0.99-1.14) 1.33 (1.19-1.49) 1.09 (0.95-1.26) 1.35 (1.26-1.45	Age, sex, and major comorbidities (diabetes, hypertension, obesity and others)
Du et al 2020, China	Retrospective cohort study	Data from Wuhan Pulmonary Hospital	December 25th, 2019 and February 7, 2020	179 hospitalized patients with COVID-19; 21 died	Hypertension	Yes vs no	4.081 (1.584–10.510)	None
Escalera-Antezana et al 2020 Bolivia	Cross-sectional study	Data from the Hospitals and the Ministry of Health of Bolivia, obtaining the clinical and epidemiological data of the COVID-19 cases that were laboratory-diagnosed	March 2 to 29, 2020	107 hospitalized patients with COVID-19; 6 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	16.33 (2.10-126.82) 3.28 (1.28-6.29) 12.13 (1.69-86.95)	Age, sex and comorbidties for only hypertension association
Esme et al 2020 Turkey	Retrospective cohort study	Data from Turkish Ministry of Health National COVID-19; aged ≥ 60 years	March 11 to May 27, 2020	16942 hospitalized patients with COVID-19; 3141 died	Diabetes Hypertension	Yes vs. no (60-79 age) Yes vs. no (80+ age)	1.18 (1.06-1.30) 1.26 (1.07-1.49) 1.17 (1.03-1.32) for	Gender, presence of hypertension, diabetes, chronic obstructive pulmonary disease (COPD), coronary artery

						Yes vs. no (60-79 age)	60-79 only	disease (CAD), atrial fibrillation (AF), chronic kidney disease (CKD), dementia, depression, malnutrition, and hyperlipidemia (HL)
Fang et al 2020 China	Retrospective cohort study	Data from admitted to the infectious ward of Wuhan Third Hospital; 615 men and 665 women, aged 7–101 years	January 27 to March 15, 2020	1280 hospitalized patients with COVID-19; 69 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.54 (1.17-10.8) 0.80 (0.35-1.83)	Sex, age, hypertension, COPD, diabetes, CRF, CHD, Arrhythmia
Farrel et al 2020, Ireland	Retrospective cohort study	Data collected from the Microbiology Department of hospital's laboratory (connolly Hospital, Ireland and Royal College of Surgeons)	March 13 and May 1, 2020	257 hospitalized patients with COVID-19; 39 died	Smoking Obesity	Yes vs. no Overweight/obese vs. no	1.73 (0.70-4.31) 3.09 (1.32-7.23)	Age
Fernandez-Cruz et al 2020, Spain	Retrospective cohort study	University hospital in Madrid, Spain	March of 2020	463 hospitalized patients with COVID-19; 71 died	Diabetes	Yes vs. no	2.80 (1.60-4.90)	None
Ferrando-Vivas et al 2020 Bolivia	Observational cohort study.	Adults from a critical care units across England, Wales, and Northern Ireland	March 1 and June 22, 2020	9,990 hospitalized patients with COVID-19; 3,933 died	BMI	per 5 kg/m2 increase	1.032 (1.003–1.061)	Several variables including age, sex, ethinicity, deprivation and others factors)
Foy et al 2020, USA	Retrospective cohort study	Data from one of 4 hospitals in the Boston, Massachusetts area (Massachusetts General Hospital, Brigham andWomen's Hospital, North Shore Medical Center, and Newton-Wellesley Hospital)	March 4 and April 28, 2020.	1641 hospitalized patients with COVID-19; 276 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.87 (0.63-1.20) 1.05 (0.80-1.4)	Age, ethnicity, D-dimer (dimerized plasmin fragment D) level, absolute lymphocyte count, hypertension, diabetes, COPD, CKD and CAD.
Fried et al 2020, USA	Retrospective cohort study	data from the hospital chargemaster, in 250 hospitals, across 38 states (patients ≥ 18 years)	February 15 and April 20, 2020	11721 hospitalized patients with COVID-19; 2514 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	0.99 (0.77-1.08) 0.78 (0.69-0.88) 1.07 (0.93-1.24) 0.72 (0.63-0.83)	Age, sex, insurance status, chronic kidney disease, liver disease obesity and smoking
Galloway et al 2020, England	Prospective cohort study	Data from two London hospitals (King's College Hospital and Princess Royal University Hospital) ; aged 71	March 1 to April 17 2020.	1157 hospitalized patients with COVID-19; 244 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.24 (0.95-1.60) 1.28 (0.99-1.67)	age and sex

		years (mean)						
Gao et al 2020, China	Retrospective cohort study	Huo Shen Shan Hospital (Wuhan)	February 5 to March 15, 2020	2877 hospitalized patients with COVID-19; 56 died	Hypertension	Yes vs. no	2.12 (1.17-3.82)	Age, sex, medical history of diabetes, insulin-treated diabetes, myocardial infarction, underwent PCI/CABG, renal failure, stroke, heart failure, and COPD
Gayam et al 2020, USA	Retrospective cohort study	Data from an inner-city teaching community hospital in New York City		408 hospitalized patients with COVID-19; 132 died	BMI	BMI per 1 unit increase	1.07 (1.04-1.11)	Age, BMI, myalgia, D-dimers, Ferritin, CRP and other biological samples
Gazzaruso et al 2020, Italy	Retrospective cohort study	Not mentionned (Italy)	15 to 30 Aril 2020	192 hospitalized patients with COVID-19; 62 died	Obesity	Yes vs no	3.62 (2.14–6.14)	Age, d-dimer, history of hypertension, eGFR, lactate dehydrogenase, history of a lung disease and C-reactive protein
Ghany et al 2020, USA	Observational cohort study	Elderly Medicare COVID-19 patients across eight states;	January 1 and May 4, 2020	400 hospitalized patients with COVID-19; 72 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI per 1 unit increase	1.03 (0.58-1.02) 1.38 (0.76-2.50) 1.00 (0.96–1.05)	Several variables (not mentionned)
Giacomelli et al 2020, Italy	Prospective cohort study	Adult COVID-19 patients admitted to Luigi Sacco Hospital in Milan,	February 21 to March 19, 2020	233 hospitalized patients with COVID-19; 48 died	Being treated with at least one anti-hypertensive agent Obesity	Yes vs. no Yes vs. no	1.41 (0.73−2.74) 3.04 (1.42−6.49)	Age, sex, comorbidities, BMI, hypertension drugs, and biological parameters
Grasselli et al 2020, Italy	Retrospective cohort study	Patients requiring ICU admission in Lombardy	February 20 to April 22, 2020.	3988 hospitalized patients with COVID-19; 926 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.18 (1.01-1.39) 0.99 (0.81-1.22)	Demographics, comorbidities, drugs, and respiratory parameters
Gu et al 2020, China	Nested case–control study	Data from mainland China outside of Hubei Province	18 December 2019 to 8 March 2020	275 hospitalized patients with COVID-19; 94 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.14 (0.70-1.90) 1.37 (0.90-2.20)	Age
Gu et al 2020, USA	Retrospective cohort study	University of Michigan Medicine Health System,	March 10 to April 22, 2020	1119 hospitalized patients with COVID-19; 43 died	Diabetes BMI Past-smoker Current-Smoker	Yes vs. no per one SD (8.1 kg/m2) Yes vs. no Yes vs. no	2.16 (0.85-5.49) 1.22 (0.74, 2.01) 2.06 (0.73, 5.77) 2.40 (0.15, 39.6)	Race, age, and sex

Guisado-Vasco et al 2020, Spain	Retrospective observational study	Data from adult patients admitted to Hospital Universitario Quironsalud Madrid, in Pozuelo de Alarcon, Madrid	March 10 to April 15, 2020	607 hospitalized patients with COVID-19; 141 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.61 (1.19-5.73) 1.73 (1.17-2.54)	Age, sex, baseline comorbidities, laboratories parameters and available therapies
Gupta et al 2020, USA	Prospective cohort study	data from multicenter cohort study of 65 hospitals, adult patients (≥18 years of age)	March 4 to April 4, 2020	2215 hospitalized patients with COVID-19; 784 died	Diabetes Hypertension current smoker BMI	Yes vs. no Yes vs. no Yes vs. no <25 25-29.9 30-34.9 35-39.9 ≥40	1.14 (0.91-1.43) 1.06 (0.83-1.36) 1.21 (0.76-1.93) 1 1.01 (0.73-1.39) 0.97 (0.69-1.37) 1.24 (0.81-1.89) 1.51 (1.01-2.25)	Age, sex, race, hypertension, BMI, diabetes, smoking and others comorbidities
Gutierrez et al 2020, Mexico	Retrospective cohort study	Data from directorate General of Epidemiology (Ministry of Health)		654,858 hospitalized patients with COVID-19; 71,706 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	1.82 (1.76-1.88) 1.39 (1.35-1.44) 1.68 (1.62–1.73) 0.97 (0.94–1.01)	Sex, age, comorbidities, health provider, whether the individual speaks an indigenous language, quintiles of the proportion of the population living in poverty in the municipality (county) of residence
Haase et al 2020, Denmark	Retrospective observational study	Data from the Danish nationwide, including 29 ICUs in 27 hospitals	10 March to 19 May 2020 and	323 hospitalized patients with COVID-19; 118 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no <18 18-24.9 25-29.9 30-34.9 35-39.9 ≥40	1.10 (0.71-1.70) 0.94 (0.64-1.37) 0.95 (0.51-1.76) 1 0.77 (0.49-1.22) 0.71 (0.39-1.27) 0.80 (0.34-1.89) 0.86 (0.38-1.93)	Age and gender
Hajifathalian et al 2020, USA	Retrospective observational study	Data from two hospitals in New York City (an academic tertiary care referral center and a smaller community hospital) (age ≥ 18)	March 4 and April 9, 2020	770 hospitalized patients with COVID-19; 88 died	BMI	<18.5 18.5-30 kg/m2 ≥30 kg/m2	1.64 (0.84-3.19) 1 1.15 (0.62-2.14)	Age, race/ethnicity, and troponin I level

Harmouch et al 2020, USA	Retrospective cohort study	Data from St. Luke's University Hospital, Bethlehem, Pennsylvania, which is part of the St. Luke's University Health Network	March and April 15, 2020	563 hospitalized patients with COVID-19; 81 died	Diabetes Hypertension Smoking BMI Obesity	Yes vs. no Yes vs. no Yes vs. no BMI per Morbid obesity vs. no	0.901 (0.547-1.482) 2.049 (1.253-3.351) 1.538 (0.935-2.530) 0.981 (0.951-1.013) 1.036 (0.587-1.829)	None
Harrison et al 2020, USA	Retrospective cohort study	Data form the TriNetX network (24 healthcare organizations in the US); aged up to 90 years	January 20 to May 26, 2020	31461 hospitalized patients with COVID-19; 1296 died	Diabetes	Yes vs. no	1.11 (0.96-1.27)	age, sex, ethnicity and comorbidities
Hashemi et al 2020, USA	Retrospective cohort study	data from nine hospitals (two large tertiary centres and seven community hospitals) and a single healthcare system in Massachusetts; aged of 63.4 years	11 March to 2 April 2020	363 hospitalized patients with COVID-19; 55 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.38 (0.68-2.79) 2.20 (0.88-5.52) 1.03 (0.51-2.09)	Age, gender, obesity, cardiac diseases, hypertension, hyperlipidaemia, and pulmonary disorders, CLD and NAFLD
Hewitt et al 2020, EUROPE	Prospective cohort study	Ten hospitals in the United Kingdom (Patients ≥ 18 years old) and one Italian Hospital (University Hospital of Modena Policlinico)	Feb 27, and April 28, 2020,	1564 hospitalized patients with COVID-19; 425 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Ex-smokers vs. never Current smokers vs. never	1.03 (0.82–1.29) 0.95 (0.77–1·18) 0.95 (0.77–1.18) 0.91 (0.59–1.42)	Age group, sex, smoking, C-reactive protein, diabetes, coronary artery disease, hypertension, renal function, and the clinical frailty scale
Hu et al 2020, China	Retrospective cohort study	Data from Renmin Hospital of Wuhan University	December 2019 to March 2020	821 hospitalized patients with COVID-19; 180 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.82 (0.34-1.97) 1.15 (0.57-2.31)	Several variables (including age, sex and comorbidities)
Iaccarino et al 2020, Italy	cross-sectional study	Data from 26 hospitals in Italy, a nationwide survey (study of the Italian Society of Hypertension)	March 9 to April 9, 2020.	1591 hospitalized patients with COVID-19; 188 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.70 (1.30-2.70) 1.00 (0.60-1.50)	Age, sex, hypertension, obesity, diabetes, HF, COPD, CKD and charlon index
Imam et al 2020, USA	Retrospective cohort study	Data from largest healthcare system in Southeast Michigan (8 hospitals);	April 1. 2020,	2040 hospitalized patients with COVID-19; 200 died	Diabetes Hypertension Smoking history Obesity	Yes vs. no Yes vs. no Yes vs. no BMI > 30 vs BMI < 30	3.90 (3.00-5.10) 4.40 (3.60-5.40) 1.40 (0.83-2.30) 1.80 (1.40-2.40)	None (multivariable for BMI only)

Ioannou et al 2020, USA	Retrospective cohort study	Data from the Department of Veterans Affairs (VA) national health care system	February 28 and May 14, 2020	10 131 hospitalized patients with COVID-19; 1094 died	Diabetes BMI Hypertension Smoking	Yes vs. no <18.5 18.5-24.9 25.0-29.9 30.0-34.9 ≥35 Yes vs. no Former vs. Never Current vs. never	1.13 (0.99-1.29) 1.29 (0.98-1.70) 1 0.90 (0.77-1.06) 0.84 (0.69-1.01) 0.97 (0.77-1.21) 0.95 (0.81-1.12) 1.02 (0.88-1.19) 0.87 (0.67-1.13)	Age, sex, race/ethnicity, sociodemographic characteristics, comorbid conditions, and symptoms
Kabarriti et al 2020, USA	Retrospective cohort study	Patients from to the Montefiore Medical Center, a large urban academic medical center in the Bronx, New York,	March 14 and April 15, 2020,	5902 hospitalized patients with COVID-19; 918 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no <18.5 18.5-35 >35	1.21 (1.06-1.40) 1.00 (0.87-1.16) 0.97 (0.66-1.41) 1 1.42 (1.19-1.70)	sex, age, socioeconomic status, ethinicity and other comorbidities
Kim et al 2020, South Korea	retrospective cohort study	Korean National Health Insurance Service Big Data linked with COVID-19 registry data of Korea Centers for Disease Control & Prevention;	confirmed COVID-19 up to March 26, 2020	9148 hospitalized patients with COVID-19; 130 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.82 (1.25-2.67) 1.32 (0.88-1.99)	age, sex, race/ethnicity, smoker, hypertension, obesity, diabetes, chronic lung disease, cardiovascular disease, neurologic disease, renal disease, immunosuppression, hematologic disorders, and rheumatologic or autoimmune disease
Kim et al 2020, USA	Retrospective cohort study	Data from 74 counties in 13 states Surveillance Network (COVID-NET); aged 62 years (median)	March 1, 2020 and May 30, 2020.	2491 hospitalized patients with COVID-19; 420 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Former smokers vs. never Current smokers	1.19 (1.01-1.40) 1.07 (0.79-1.45) 1.14 (0.98-1.33) 1.22 (0.99-1.51)	Sex, age, type of distiricts, high epidemic region and socio-economic status

					Obesity	BMI≥30 vs BMI<30	1.09 (0.92-1.30)	
Klang et al 2020, USA	Retrospective cohort study	Data from five hospital campuses serving different geographic populations (Mount Sinai Hospital, Mount Sinai Brooklyn, Mount Sinai Queens, Mount Sinai Morningside, and Mount Sinai West), in NY	March 1 and 5/9/2020	6760 hospitalized patients with COVID-19; 1320 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Smokers (past or current) vs. never	1.00 (0.80-1.30) 1.20 (0.90-1.60) 2.30 (1.80-3.00) 0.70 (0.50–0.90)	Age decile, CAD, DM, CHF, CKD, COPD, obesity, cancer, and smoking
Klang et al 2020, USA	Retrospective cohort study	Data of COVID-19 patients hospitalized to a large academic hospital system in New York City	March 1 and May 17, 2020	3406 hospitalized patients with COVID-19; 1136 died	Age ≤ 50 Diabetes Hypertension Smoking BMI Age > 50 Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no past or present vs. never 30 30-40 >40 Yes vs. no Yes vs. no Smokers (past or present) vs. never 30 30-40 >40	1.30 (0.70-2.60) 0.50 (0.20-1.10) 1.70 (0.80-3.80) 1 1.10 (0.50-2.30) 5.10 (2.30-11.10) 1.4 (1.2-1.7) 1.1 (0.9-1.3) 1.0 (0.8-1.2) 1 1.1 (0.9-1.3) 1.6 (1.2-2.3)	Age, sex, CAD, CHF, hypertension, CKD, smoking, BMI, race, cancer and hyperlipidemia
Kolhe et al 2020, UK	Retrospective cohort study	Data from the Department of University Hospitals of Derby and Burton (UHDB) National Health Service (NHS), which comprised 5 hospitals in the UK.	March 5 and May 12, 2020	1161 hospitalized patients with COVID-19; 419 died	Diabetes with complications	Yes vs. no	1.15 (0.82, 1.61)	Age, gender, ethnicity, residence in care home, use of ACEI/ARB, need for mechanical ventilation, and comorbidities
Lampasona et al 2020, Italy	Prospective cohort study	Data from the Emergency or Clinical departments of the Istituto di Ricovero e Cura a Carattere Scientifico;	480 February and 19 April 2020	509 hospitalized patients with COVID-19; 93 died	Diabetes	Yes vs. no	2.32 (1.44-3.75)	Age, sex, BMI and other relevant comorbidities
Lee et al 2020,	Retrospective	Patients in the	February 18 and	98 hospitalized patients	Diabetes	Yes vs. no	4.74 (1.68-13.38)	None

South Korea	cohort study	Daegu/Gyeongsangbuk-do province; patients aged ≥ 65 years	March 4, 2020	with COVID-19;20 died	Hypertension	Yes vs. no	1.95 (0.70-5.42)	
Li et al 2020, China	Retrospective cohort study	Wuhan Union Hospital Affiliated with Tongji Medical College of Huazhong University of Science	22 January 2020 to 17 March 2020.	453 hospitalized patients with COVID-19; 39 died	Diabetes Hypertension	known diabetes vs. normal glucose Newly diagnosed diabetes vs. normal glucose	4.63 (1.02-21.00) 9.42 (2.18-40.7)	Age, sex, smoking, systolic blood pressure, total cholesterol, anti-hypertensive drugs and admission to ICU.
Li et al 2020, China	Retrospective cohort study	Patients admitted to Renmin Hospital of Wuhan University	January 31 to February 20, 2020	204 hospitalized patients with COVID-19; 76 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.60-1.90) 2.30 (1.50-3.60)	Age
Li et al 2020, China	Retrospective cohort study	Patients from two critical and two general designated hospitals, in Wuhan	January 2 and February 15, 2020	523 hospitalized patients with COVID-19; 95 died	BMI	per 2.9 kg/m2 increase	0.99 (0.90-1.10)	age, gender, Hypertension, diabetes mellitus, coronary artery disease, þCr, PCT, ALC, cTnI, hs-CRP, LDL-c and FBG
Li et al 2020, China	Retrospective cohort study	Patients admitted to Wuhan Red Cross Hospital	January 24 and March 15, 2020	199 hospitalized patients with COVID-19; 18 died	Diabetes Hypertension	Yes vs. no Yes vs. no	10.816 (1.895-61.741) 0.660 (0.150-2.908)	Multivariate model but variable are not mentionned
Li et al 2020, USA	Retrospective cohort study	Yale New Haven Health System, a 5-hospital system in the Northeast of the United States	February 15 and April 25, 2020	1685 hospitalized patients with COVID-19; 318 died	Diabetes wit complication s	Yes vs. no	1.00 (0.70-1.40)	Age, sex, race/ethnicity, medical comorbidities and hospital locations
Liu et al 2020, China	Retrospective cohort study	Leishenshan Hospital	NA	934 hospitalized patients with COVID-19; 13 died	Diabetes	Yes vs. no	2.14 (0.58-7.91)	Age, the history of cardiovascular disease, D-dimer, WBC, and lymphocyte count
Liu et al 2020, China	Retrospective cohort study	Wuhan Infectious Disease Hospital (18–94 years)	December 29, 2019 to February 28, 2020	1190 hospitalized patients with COVID-19; 157 died	Diabetes	Yes vs. no	11.16 (1.87-66.57)	Age, sex, diabetes, hypertension, CKD and some symptoms and signs
Liu et al 2020, China	Retrospective cohort study	Weibo posts and a telephone survey	January 20 and February 15, 2020,	599 hospitalized patients with COVID-19; 83 died	Hypertension Smoking	Yes vs. no Yes vs. no	0.24 (0.08-0.67) 0.90 (0.44-1.85)	Age and sex
Liu et al 2020, China	Retrospective cohort study	Patients of Eastern Campus of Renmin Hospital, Wuhan University	January 30 to February 20, 2020	214 hospitalized patients with COVID-19; 94 died	Hypertension	Yes vs. no	0.82 (0.40-1.70)	Age, lymphocytopenia, serum LDH, PCT, and BUN levels
Lombardi et al	Cross-sectional	13 Italian cardiology units	March 1 to April	614 hospitalized patients	Diabetes		1.41 (0.99-2.02)	For hypertension :

2020, Italy	study		9, 2020	with COVID-19; 148 died	Hypertension Obesity BMI Smoking	Yes vs. no Yes vs. no Yes vs. no +1 kg/m2 Ever vs. never smoker	1.26 (0.81-1.96) 1.26 (0.85-1.85) 1.01 (0.97-1.04) 1.33 (1.24-1.44)	ajusted for comorbidities
Lopez-Mendez et al 2020, Mexico	Retrospective cohort study	Patients admitted to Medica Sur Clinic & Foundation	March 14 and June 5, 2020	155 hospitalized patients with COVID-19; 11 died	Hypertension	Yes vs. no	3.96 (1.28−12.19)	None
Lunski et al 2020, USA	Retrospective cohort study	Data from ochsner Health System in the state of Louisiana,	March 1 through April 30, 2020	4760 hospitalized patients with COVID-19; 484 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	0.92 (0.70-1.21) 0.60 (0.47-0.78) 1.30 (1.03-1.63)	Age, sex, race, comorbidities (COPD, diabetes, BMI, hypertension, CKD) and other variables
Maddaloni et al 2020, Italy	Retrospective cohort study	Data from four academic hospitals located in the Lazio region of Italy (the Covid-19 & Diabetes (CoViDiab))	March 1 to May 15th, 2020	354 hospitalized patients with COVID-19; 77 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Ever vs. never smoker	1.30 (0.50-3.10) 2.80 (1.00-7.50) 1.00 (0.40-2.70)	Age and sex
Magleby et al 2020, USA	Retrospective cohort study	Hospitalized patients in NewYork-Presbyterian Hospital/Weill Cornell Medical Center and affiliated Lower Manhattan Hospital	March 30 to April 30, 2020	678 hospitalized patients with COVID-19; 127 died	Hypertension	Yes vs. no	0.74 (0.41-1.33)	Age, race, coronary artery disease, congestive heart failure, cerebrovascular disease, hypertension, COPD, days of symptoms prior to admission, symptoms upon presentation, initial chest x-ray findings, and level of oxygen support within three hours of arrival to the ED
Mallow et al 2020, USA	Retrospective cohort study	Data were obtained retrospectively from a commercially-available, allpayer administrative database of inpatient and hospital-based outpatient (276 hospitals in the database	March 15 and April 30, 2020	21 676 hospitalized patients with COVID-19; 4936 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.30 (1.17-1.44) 0.77 (0.70-0.86) 1.30 (1.15 1.47)	Geographic region, teaching status and bed size
Mather et al	Retrospective	Data from Hartford Hospital,	February 24 and	878 hospitalized patients	Obesity	Yes vs no	5.87 (2.54-13.6)	Age, history of atrial

2020, USA	cohort study	an 890-bed tertiary care medical center in Hartford, Connecticut	May 13, 2020	with COVID-19; 191 died				fibrillation, heart failure, CKD, obesity, neutrophil-lymphocyte ratio, white blood cell count, and asthma
Matsuzawa et al 2020, Japan	Retrospective cohort study	Patients from six hospitals in Kanagawa Prefecture	February 1 and May 1, 2020	151 hospitalized patients with COVID-19; 14 died	Hypertension	Yes vs. no	0.63 (0.15-2.34)	Age, sex, and diabetes mellitus
McCarty et al 2020, USA	Retrospective cohort study	Data from two tertiary medical centers and the seven community hospitals which comprise Mass General Brigham	March 22, through April 2, 2020	379 hospitalized patients with COVID-19; 57 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.97 (0.97-4.02) 2.15 (0.94-4.91) 0.69 (0.35-1.37)	Age, gender, obesity, cardiopulmonary comorbidities, hypertension, and diabetes
MCCULLOUGH et al 2020, USA	Retrospective observational cohort	Data from weill Cornell Medicine/New York-Presbyterian Hospital, a quaternary referral center and 862-bed teaching hospital	March 3 through April 9, 2020	756 hospitalized patients with COVID-19; 90 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Former vs. never Active vs. never	1.10 (0.68-1.70) 2.48 (1.51-4.07) 0.70 (0.43-1.14) 2.26 (1.42-3.61) 0.33 (0.04-2.47)	Age, clinical characteristics, and ECG variables, CAD, an immunosuppressed state, and hypoxemia
Mendy et al 2020, USA	Retrospective observational cohort	Data from university of Cincinnati health system, consisted of 4 hospitals in the states of Ohio. Kentucky. and Indiana	March 13 to May 31, 2020	689 hospitalized patients with COVID-19; 26 died	Diabetes Smoking Obesity	Yes vs. no Ever vs. never smokers Yes vs. no	1.79 (0.75-4.27) 0.82 (0.33-2.01) 1.09 (0.36-3.33)	Age, gender, race/ethnicity, and smoking
Menezes-Soares et al 2020, Brazil	Retrospective cohort study	Data from State Health Secretariat of Espırito Santo; Brazil	February 29 to June 11th, 2020	1,152 hospitalized patients with COVID-19; 456 died	Diabetes	Yes vs. no	1.88 (1.43-2.47)	None
Meng et al 2020, China	Retrospective cohort study	Data from Tongji Hospital in Wuhan	January 18 and March 27, 2020	2665 hospitalized patients with COVID-19; 299 died	Diabetes Hypertension	Yes vs. no Yes vs. no	0.71 (0.48-1.03) 1.01 (0.76-1.35)	Age, sex and comorbidities (diabetes, hypertension, COPD, CKD, CVD and cancer)
Mikami et al 2020, USA	Retrospective cohort study	Data from the medical records of the Mount Sinai Health System (8 hospitals and/or over 400 ambulatory practices)	March 13 and April 17, 2020	3708 hospitalized patients with COVID-19; 806 died	Diabetes Hypertension Cigarettes use history	Yes vs. no Yes vs. no Yes vs. no	0.92 (0.73-1.16) 0.91 (0.79-1.07) 1.01 (0.90–1.13)	Age, sex, race, cigarette use history, past medical history of asthma, hypertension, diabetes, or

		in the New York City metropolitan area						cancer, systolic BP, RR, SpO2, BMI, initial laboratory values (lymphocyte proportion, Ddimer IL-6), and hydroxychloroquine use
Mirani et al 2020, Italy	large case series	Patients from single center in the region of Lombardy	February 20 and 9 April 2020	385 hospitalized patients with COVID-19; 102 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no BMI>30 kg/m2	1.33 (0.88-2.00) 1.84 (1.15–2.95) 1.33 (0.81–2.18)	Age and sex
Moon et al 2020, South Korea	Retrospective cohort Study	Data identified from the Korean Health Insurance Review and Assessment Service	prior to May 15. 2020.	5307 hospitalized patients with COVID-19; 211 died	Diabetes	Yes vs. no	2.66 (1.90-3.73	Age, sex, hypertension, dyslipidemia, chronic kidney disease, chronic obstructive lung disease, cardiovascular disease, atrial fibrillation, end-stage renal disease and cancer
Moon et al 2020, South Korea	Retrospective cohort study	Data from Dongguk University Gyeongju Hospital (DUGH) and the Andong Medical Center (AMC)	February 18 and June 30, 2020	352 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.69 (1.05-6.85) 1.90 (0.75-4.84)	Age
MOWLA et al 2020, Bangladesh	Cross-sectional Study	Patients admitted in Dhaka Medical College Hospital (DMCH)	May 2 and 15, 2020	100 hospitalized patients with COVID-19; 10 died	Hypertension Smoking	Yes vs no Yes vs no	2.89 (0.54-15.37) 9.95 (1.73-57.12)	Multiple variables (hypertension, heart disease and renal disease)
Munblit et al 2020, Russia	Retrospective cohort study	Patients from the Sechenov University hospital network in Moscow city	April 22 and June 6, 2020	3480 hospitalized patients with COVID-19; 191 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.10 (1.46-2.99) 1.70 (1.14-2.59)	Age, sex, hypertension, diabetes, dementia, neoplasm, COPD and CKD
Munoz et al 2020, Spain	Retrospective cohort study	Patients a large hospital in Madrid,	NA	100 hospitalized patients with COVID-19; 23 died	Hypertension	Yes vs. no	4.47 (1.36-14.66)	Age, BMI, diabetes and antiobiotic treatment
Munoz-price et al 2020, USA	cross-sectional study	Froedtert Health and Medical College of Wisconsin (Milwaukee), the largest academic system in Wisconsin,	March 12 to March 31, 2020	369 hospitalized patients with COVID-19; 20 died	Smoking Obesity	History of smoking or current Yes vs. no	7.15 (0.55-9.22) 1.19 (1.05-1.35)	clustering at the zip code–level of patients' residence
Nachega et al 2020, Congo	Retrospective cohort study	Patients from seven hospitals in Kinshasa (one private, two	March 10, and July 31 2020	766 hospitalized patients with COVID-19; 101	Diabetes Hypertension	Yes vs. no Yes vs. no	1.10 (0.66-1.81) 1.00 (0.62-1.61)	age-group, the presence of comorbidities

		faith-based Catholic, and four public)		died	Obesity	Yes vs. no	2.30 (1.24–4.27)	(hypertension, heart disease, diabetes, obesity, CKD, and cancer) and treatment received
Nakeshbandi et al 2020, USA	Retrospective cohort study	Patients tested for COVID-19 at SUNY Downstate Health Sciences University, hospital in New York	March 10 and April 13, 2020	504 hospitalized patients with COVID-19; 219 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Overweight Obese	1.10 (0.92-1.30) 0.81 (0.66–1.0) 1.4 (1.1–1.9) 1.3 (1.0–1.7)	Age, sex, smoking, obesity and hypertension
Namendys-Silva et al 2020, Mexico	Retrospective cohort study	Data from 10 hospitals located in Mexico	April 1 and April 30, 2020	164 hospitalized patients with COVID-19; 85 died	Diabetes Hypertension	Yes vs. no Yes vs. no	3.03 (1.51-6.09) 2.41 (1.26-4.62)	None
Narain et al 2020, USA	Retrospective cohort study	Data from electronic health records across the Northwell Health system	March 1 and April 24, 2020	5,776 hospitalized patients with COVID-19; 282 died	Diabetes Hypertension Smoking BMI	Yes vs. no Yes vs. no Never Active Former Smoker (unknown active/former) Unknown < 18.5 18.5-24.9 25-29.9 >30	1.02 (0.89-1.17) 0.83 (0.73-0.94) 1 1.43 (0.94-2.21) 0.93 (0.78-1.11) 1.42 (1.09-1.83) 3.02 (2.58-3.56) 1.15 (0.93-1.44) 1 0.98 (0.83-1.42) 1.07 (0.90-1.27)	Several covariables not mentionned
Neumann-Podczaska et al 2020, Poland	Retrospective cohort study	Data from the Jozef Strus Hospital, Poznan, Poland	March 12 and 5 April 2020	50 hospitalized patients with COVID-19; 20 died	Diabetes	Yes vs. no	2.35 (0.97–5.67)	Age
Nicholsoni et al 2020, USA	Retrospective cohort study	Patients admitted in five hospitals in the Mass General Brigham health care system, Boston	COVID-19 before May 19, 2020.	1042 hospitalized patients with COVID-19; 211 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Former vs. no Current vs. no	2.32 (1.30-4.12) 1.81 (1.32-2.49) 2.05 (1.39-3.03) 1.06 (0.78-1.44)	Age, sex, statin, C-reactive protein, albumin, C-reactive protein, diabetes, neutrophil-lymphocyte ratio, mean corpuscular volume, platelet count, and procalcitonin

Nie et al 2020, China	Retrospective cohort study	Patients admitted to Union Hospital (West Campus), Huazhong University of Science and Technology	January 12 and March 12, 2020	311 hospitalized patients with COVID-19; 111 died	Diabetes Hypertension smoking	Yes vs. no Yes vs. no Current vs. never	2.13 (1.15–3.95) 3.80 (2.31–6.25) 3.66 (1.41–9.46)	None
Nogueira et al 2020, Portugal	Retrospective cohort study	Data from the General Health Directorate of Portugal (DGS)	January to April, 21 2020	20293 hospitalized patients with COVID-19; 502 died	Diabetes	Yes vs. no	0.75 (0.55-1.01)	Age, sex and pregnancy
Núñez-Gil et al 2020, Spain	Retrospective cohort study	Data from the HOPE COVID-19 registry (Health Outcome Predictive Evaluation for COVID-19)	February 8 and April 1, 2020	908 hospitalized patients with COVID-19; 301 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. exsmoker No vs. exsmoker	2.77 (1.96-3.90) 1.74 (1.01-2.99) 1.52 (0.83-2.76) 1.07 (0.54-2.11) 0.55 (0.39-0.80)	None
Palaiodimos et al 2020, USA	Retrospective cohort study	Conducted at the Montefiore Medical Center, a tertiary academic institution in the Bronx, New York	43899	200 hospitalized patients with COVID-19; 48 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no <25 BMI ≥35 BMI Current or former smoker vs. never	1.16 (0.55-2.44) 0.93 (0.44–1.98) 1.57 (0.68–3.66) 3.35 (1.43–7.87) 0.83 (0.37–1.87)	Age, BMI, smoking and diabetes
Pan et al 2020, China	Retrospective cohort study	Renmin Hospital of Wuhan University	January 4, to February 14, 2020	996 hospitalized patients with COVID-19; 89 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.34 (0.78-2.29) 2.24 (1.36–3.70)	Age, sex, diabetes mellitus, matched with hypertensive
Park et al 2020, South Korea	Retrospective cohort study	Patients from five tertiary hospitals of Daegu, South Korea; aged >19 years	February 20 and April 14, 2020	289 hospitalized patients with COVID-19; 70 died	Diabetes	Yes vs. no	2.226 (1.357 to 3.652)	Age, sex, comorbidities (diabetes, hypertension, COPD, CKD and CLD) and other biological parameters
Parker et al 2020, South Africa	Retrospective cohort study	Patients admitted to the Tygerberg Hospital, Cape Town	25 March 25 and May 11, 2020	113 hospitalized patients with COVID-19; 28 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.90 (0.80-4.60) 2.40 (1.00-5.80) 1.30 (0.50-3.20)	None

Perkin et al 2020, England	Retrospective cohort study	Data for deaths occurring at St George's HOSPITAL (Black, Asian and minority ethnic (BAME) communities), London	6 weeks commencing 12th of March 2020	243 hospitalized patients with COVID-19 and died with it	Diabetes Hypertension	Yes vs. no Yes vs. no	1.76 (1.05-2.94) 1.40 (0.80-1.99)	Age
Petrilli et al 2020, USA	Prospective cohort study.	Data from 260 outpatient office sites and four acute care hospitals (Tisch Hospital and NYU Langone Orthopedic Hospital in Manhattan. NYU Langone Hospital – Brooklyn in Brooklyn. and NYU Winthrop on Long Island).	March 1 to April 8 2020	2737 hospitalized patients with COVID-19; 391 died	Diabetes Hypertension BMI Smoking	Yes vs. no Yes vs. no <25 25.0-29.9 30.0-39.9 ≥40 Former vs no Current vs no Unknown vs no	1.01 (0.85-1.21) 0.98 (0.78-1.23) 1 1.01 (0.82-1.25) 1.08 (0.87-1.36) 1.45 (0.99-2.13) 1.07 (0.88-1.31) 0.92 (0.62-1.38) 1.50 (1.21-1.86)	Age, sex, race/ethnicity, smoking status, BMI, and chronic diseases
Pettit et al 2020, USA	Retrospective cohort study	Patients admitted to the University of Chicago Medical Center, an 811-bed academic medical center on the south side of Chicago, who had completed their hospital course	March 1 and April 18, 2020	238 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	0.50 (0.20-1.70) 0.30 (0.10-0.90) 1.70 (1.10-2.80)	Demographics variable (age, sex, race/ethnicity) and comorbidities (diabetes, hypertension, pulmonary diseases, CVD, cancer, HIV and stroke)
Poblador-Plou et al 2020, Spain	Retrospective cohort study	The PRECOVID Study; in the Spanish region of Aragon	March 4 and April 17, 2020	4412 hospitalized patients with COVID-19; 771 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.65 (1.23-2.21), men 1.40 (1.02-1.91), women 0.96 (0.75-1.23), men 1.20 (0.92-1.56), women	Age
Portoles et al 2020, Spain	Prospective cohort study.	Patients form Puerta de Hierro University Hospital	February 25 to 24 April, 2020	1603 hospitalized patients with COVID-19; 197 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.83 (1.34-2.50) 1.22 (0.63–2.33)	Age
Price-Haywood et al 2020, USA	Retrospective cohort study	Data from patients seen within an integrated-delivery health system (Ochsner Health) in Louisiana	March 1 and April 11, 2020	3481 hospitalized patients with COVID-19; 326 died	Obesity	Yes vs. no	0.99 (0.77–1.27)	Age, sex, race, Charlson Comorbidity Index score, residence in a low-income area, and indicators for baseline vital signs and laboratory measures (respiratory

								rate; levels of aspartate aminotransferase, venous lactate, creatinine, bilirubin, procalcitonin, and C-reactive protein; and counts of lymphocytes and platelets).
Qin et al 2020, China	Retrospective cohort study	Patients admitted to the Sino-French New City Branch of Tongji Hospital, Tongji Medical college, Huazhong University of Science and Technology	January 26 and February 5, 2020	548 hospitalized patients with COVID-19; 90 died	Smoking	Smoking history vs. no	0.92 (0.53-1.57)	Age, sex, and comorbidities
Ramos-Rincon et al 2020, Spain	Retrospective cohort study	Data from the Spanish Society of Internal Medicine's SEMI-COVID-19 Registry, in which 150 hospitals in Spain participate (patients ≥18 years-old of age)	March 1 and May 29, 2020	2772 hospitalized patients with COVID-19; 1301 died	Obesity	BMI ≥ 30 vs. BMI < 30	1.24 (1.00-1.54)	None
Rapp et al 2020, USA	Retrospective cohort study	Data from the Mount Sinai Hospital System (MSHS) COVID-19 registry	February 29 and May 19, 2020	4,062 hospitalized patients with COVID-19; 1,190 died	Obesity	BMI ≥ 35 vs. BMI < 35	1.53 (1.21-1.94)	Sex, age and history of CKD
Rastad et al 2020, Iran	Retrospective cohort study	Alborz province, Iran	February 20 to March 25, 2020	2957 hospitalized patients with COVID-19; 301 died	Diabetes	Yes vs. no	1.69 (1.05-2.74)	Gender, age and laboratory tests
Recinella et al 2020, Italy	Retrospective cohort study	Patients admitted to two COVID-19 units of Sant'Orsola-Malpighi University Hospital in Bologna; age ≥ 65	March 30 and May 15, 2020	109 hospitalized patients with COVID-19; 43 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI per	0.751 (0.347–1.622) 1.006 (0.495–2.047) 0.875 (0.782–0.979)	Age
Reilev et al 2020, Denmark	Retrospective cohort study	Data from the Danish Microbiology Database	February 27 until 19 May 2020	11 122 hospitalized patients with COVID-19; 577 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no overweight and obesity vs. no	1.60 (1.30–2.00) 1.30 (1.10-1.60) 1.50 (1.10-2.00)	Age and sex
Rivera-Izquierdo et al 2020, Spain	Retrospective cohort study	Patients from Hospital Universitario Clı´nico San Cecilio (Granada, Spain); aged 24 to 97	16 March and 10 April 2020	238 hospitalized patients with COVID-19; 38 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.42 (1.43-4.09) 1.48 (0.80-2.71)	Age, sex, comorbidities and medication on admission, symptoms, physical exam on

								admission
Rodrıguez-Molinero et al 2020, Spain	Retrospective cohort study	Patients from the hospitals of the Consorci Sanitari de l'Alt Penedès i Garraf (CSAPG).	March 12 and May 2 2020	418 hospitalized patients with COVID-19;79 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no Yes vs. no Yes vs. no	2.96 (1.75-4.99) 5.64 (3.13-10.87) 1.12 (0.58–2.06) 0.38 (0.09–1.11)	None
Rosenberg et al 2020, USA	Retrospective cohort study	Patients admitted to hospitals in the New York City (NYC) metropolitan region between	March 15 and 28, 2020.	1438 hospitalized patients with COVID-19; 264 died	Diabetes Hypertension Obesity Smoking	Yes vs. no Yes vs. no BMI >30 vs BMI <30 kg/m2 Current vs. never Farmer vs. never	1.57 (1.28-1.93) 1.95 (1.55-2.46) 0.96 (0.75-1.22) 1.30 (0.74-2.29) 1.47 (1.11-1.95)	Age
Rossi et al 2020, Italy	Prospective cohort	Patients from the Reggio Emilia province (Hospital, outpatient, primary, and preventive care to all the resident population)	February 27 to April 2. 2020	2653 hospitalized patients with COVID-19; 217 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.60 (1.10-2.20) 1.60 (1.20-2.10) 1.30 (0.60-2.90)	Age and sex and Charlson Comorbidity Index
Rottoli et al 2020, Italy	Retrospective cohort study	Data of the ongoing PREDI-CO multicentric study, from Sant'Orsola Hospital in Bologna (≥18 years)	March 1 and April 20. 2020.	482 hospitalized patients with COVID-19; 31 died	Diabetes Hypertension Obesity BMI	Yes vs. no Yes vs. no Yes vs. no per 1 unit increase <30 30-35 ≥35	1.05 (0.64-1.74) 1.23 (0.74-2.04) 2.18 (1.34–3.54) 1.07 (1.02–1.13) 1 1.21 (0.64–2.27) 1.72 (1.00–2.99)	Age, gender, hypertension and type 2 diabetes
Rozaliyani et al 2020, Indonesia	Retrospective cohort	Data from the ongoing recapitulation of Epidemiological Surveillance (ES) by the Provincial Health Office of Jakarta	March 2 and to April 27th 2020	4052 hospitalized patients with COVID-19; 381 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.26 (0.80-1.98) 1.86 (1.24-2.78)	Age, sex, symptoms, and other comorbidities
Russo et al 2020, Italy	Retrospective cohort study	Patients admitted to the Policlinico San Martino Hospital, Genoa and aged ≥ 18 years	February 25 and April 13, 2020	777 hospitalized patients with COVID-19; 275 died	Hypertension	Yes vs. no	0.91 (0.61–1.34)	Age, sex, Charlson morbidity and others biological informations
Salacup et al 2020, USA	Retrospective cohort study	A Single Tertiary Center Cohort in an underserved-Inner City Population with minority population in the United States	March 1 to April 24,2020	242 hospitalized patients with COVID-19; 52 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no BMI	1.45 (0.67-3.13) 0.95 (0.38-2.43) 1.02 (0.98-1.07)	Age, BMI, race/ethinicty and comorbidities

Salazar et al 2020, USA	Retrospective cohort study	Data from the eight Houston Methodist hospitals	March 28 to July 6, 2020	316 hospitalized patients with COVID-19; 24 died	Diabetes Hypertension Obesity BMI	Yes vs. no Yes vs. no <30 ≥30 <18.5 18.5-24.9 25-29.9 ≥30 BMI (unit)	5.33 (2.00-14.18) 5.60 (1.88-16.63) 1 0.78 (0.36, 1.69) 201.94 (28.85, 1413.70) 1 0.40 (0.11, 1.39) 0.49 (0.18, 1.33) 1.02 (0.95, 1.09)	Age
Sands et al 2020, USA	Retrospective cohort study	Data from HCA Healthcare, a large healthcare system with >180 affiliated acute-care facilities among its >2,000 associated sites of care in 21 US states and the United Kingdom.	March 1 and discharge dates as of May 12. 2020.	6,180 hospitalized patients with COVID-19; 1,181 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.57 (1.21-2.03) 0.63 (0.48-0.82) 1.41 (1.08-1.85)	Age, sex, all comorbidities, vital signs at admission, and other patient characteristics
Santos et al 2020, Brazil	Retrospective cohort study	Data registered in the public national epidemiological surveillance system and obtained from the open access database (3103 health units and 1238 municipalities)	20 February 2020 to 2 June 2020	46 285 hospitalized patients with COVID-19; 21408 died	Diabetes BMI	Yes vs. no 18.5-24.9 25-29.9 30-34.9 35-40 >40	1.19 (1.14–1.23) 1 1.18 (0.53–2.61) 1.00 (0.47–2.12) 1.16 (0.54–2.45) 1.27 (0.59–2.70)	Age, sex, race/ethnicity,, education level, symptoms and signs and comorbidities
Seiglie et al 2020, USA	Retrospective cohort study	Data from the Massachusetts General Hospital, registry of patients hospitalized	March 11 and April 30, 2020	417 hospitalized patients with COVID-19; 49 died	Diabetes Hypertension BMI	Yes vs. no Yes vs. no 25-29.9 ≥30	2.02 (1.01-4.03) 1.33 (0.57-3.10) 0.60 (0.25-1.44) 1.10 (0.50-2.45)	Age, sex, race/ethnicity, and comorbidities
Shah et al 2020, USA	Retrospective cohort study	Patients from Phoebe Putney Health System, the largest community health system in Southwest Georgia	March 2 and May 6, 2020.	522 hospitalized patients with COVID-19; 92 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no	1.51 (0.90-2.55) 3.36 (1.30-8.60) 1.03 (0.55-1.93)	Age, BMI, sex, race and comorbidities

					BMI	Former/current vs. never smoker <=30 30-40 >=40	1 1.49 (0.79-2.77) 2.29 (1.11-4.69)	
Shang et al 2020, China	Retrospective cohort study	Data from the Wuhan No.7 Hospital; temporarily taken over by Zhongnan Hospital of Wuhan University.	December 25, 2019 to March 20, 2020	584 hospitalized patients with COVID-19; 57 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.18 1.07-4.44) 0.70 (0.36-1.35)	Age, sex, hypertension, cardiovascular diseases, diabetes, chronic respiratory diseases, chronic kidney diseases, chronic liver diseases, acute kidney injury, acute liver injury, respiratory failure, and acute cardiac injury
Shi et al 2020, China	Retrospective cohort study	Patients from two tertiary hospitals in Wuhan (Renmin Hospital of Wuhan University and Zhongnan Hospital of Wuhan University)	January 1 and March 8, 2020	1561 hospitalized patients with COVID-19; 31 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.84-2.99) 2.50 (1.30-4.78)	Age and sex
Shi et al 2020, China	Retrospective cohort study	Patients admitted to the Renmin Hospital of Wuhan University with laboratory-confirmed COVID-19	January 1 and February 23, 2020	671 hospitalized patients with COVID-19; 62 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.16 (0.47-2.56) 1.07 (0.46-2.53)	Age, preexisting cardiovascular disease (hypertension and other), diabetes, cerebrovascular disease, COPD and ARDS
Smith et al 2020, USA	retrospective cohort study	Data from four Trinity Health of New England hospitals (THONE) in Connecticut and Massachusetts	March 1 to April 22, 2020	346 hospitalized patients with COVID-19; 117 died	Diabetes	Yes vs. no	1.43 (1.07-1.92)	Age, COPD and hyperlipidemia
Sousa et al 2020, Brazil	Historical cohort	Datat from Fortaleza (Ceará's capital city), a state of the Brazilian Northeast	until 14 April 2020	2070 hospitalized patients with COVID-19; 131 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.30 (0.80-2.10) 1.70 (0.40-7.70)	Age, cardiovascular disease, diabetes, BMI, Neurologic disease, pneumopathis and renal disease.
Steinberg et al 2020, USA	retrospective cohort study	Patients two sites: a high-volume, urban, academic,	March 8–April 4, 2020	210 hospitalized patients with COVID-19; 18 died	Obesity	Yes vs no	6.29 (1.76-22.46)	None

		tertiary-care medical center and an affiliated suburban community hospital; 18-45 years old						
Stony Brook COVID-19 Research Consortium et al 2020, USA	Retrospective cohort study	Data of 1325 sequentially hospitalized patients with COVID-19 in New York;177 died; aged 49 to 75	March 2 to May 11, 2020	1325 hospitalized patients with COVID-19; 177 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no BMI >30 vs BMI <30 kg/m2	1.78 (1.21-2.65) 3.24 (1.98-5.27) 0.93 (0.63-1.37)	None
Sun et al 2020, China	Retrospective cohort study	Data of the Sino-French New City Branch of Tongji Hospital,	January 29 and March 5, 2020	244 hospitalized patients with COVID-19; 121 died	Hypertension	Yes vs no	0.82 (0.24-2.75)	Age, sex, biological parameter, symptoms and signs
Talavera et al 2020, Spain	Retrospective cohort study	Patients from Hospital Clínico Universitario de Valladolid	March 8 to April 11, 2020	576 hospitalized patients with COVID-19; 127 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Current or recent vs. never	1.33 (0.76-2.34) 1.31 (0.76-2.28) 1.71 (0.93-3.12)	Age, sex, hypertention, anosmia, cardiological disorders, cancer, smoking and Prior mRS ≥ 3
Tartof et al 2020, USA	Retrospective cohort study	Data from Kaiser Permanente Southern California members, a large integrated health care organization	February 13 to May 2 2020	6916 hospitalized patients with COVID-19; 206 died	Diabetes Smoking Hypertension BMI	DM with hemoglobin A1c level <7.5% vs. no DM with hemoglobin A1c level >7.5% vs. no Current Former Never Yes vs. no <18.5 18.5–24 25–29 30–34 35–39 40–44 ≥45	1.12 (0.79-1.59) 1.22 (0.81-1.84) 0.84 (0.30–2.35) 1.30 (0.96–1.75) 1 1.38 (0.93–2.06) 1.81 (0.99–3.30) 1 0.91 (0.62–1.35) 1.26 (0.82–1.95) 1.16 (0.63–2.17) 2.68 (1.43–5.04) 4.18 (2.12–8.26)	Sex, race/ethnicity, age, tobacco use, myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, chronic pulmonary disease, renal disease, metastatic tumor or malignancy, other immune disease, hyperlipidemia, hypertension, asthma, organ transplant, and diabetes status and hemoglobin A1c level.
Tehrani et al 2020, Sweden	Retrospective cohort study	Patients admitted to the Danderyd University Hospital	March 5 and April 28, 2020	255 hospitalized patients with COVID-19; 70 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.35 (0.68–2.66) 1.37 (0.77–2.46) 1.07 (0.61–1.87)	None

Turcotte et al 2020, USA	Retrospective cohort study	Data from anne Arundel Medical Center, a 350-bed acute care hospital	March 1 to April 12, 2020	117 hospitalized patients with COVID-19; 29 died	Diabetes	Yes vs. no	3.12 (0.75-12.97)	Temperature at admission, supplemental O2, sputum production and chronic kidney diseases
Wang et al 2020, China	Retrospective cohort study	Data from the intensive care unit of the Caidian Branch of Tongji Hospital	February 2 AND March 1, 2020	59 hospitalized patients with COVID-19; 41 died	Smoking	Smoking history vs. no	1.57 (0.74–3.31)	Age
Wang et al 2020, China	Retrospective cohort study	Patients admitted to Wuhan Union West Hospital and Wuhan Red Cross Hospital;	January 24 to February 10, 2020	605 hospitalized patients with COVID-19; 114 died	Hypertension	Yes vs. no	1.27 (0.85-1.89)	Age
Wang et al 2020, China	Retrospective cohort study	Data from Several wuhan hospitals	February 23; 2020	7283 hospitalized patients with COVID-19; 649 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.67 (1.22-2.30) 5.58 (4.30-7.26)	age, sex, location, occupation, dichotomous self-reported co-morbidities, number of presenting symptoms
Wang et al 2020, China	Retrospective cohort study	Patients hospitalized in the First People's Hospital of Jiangxia District in Wuhan, China	January 7, to February 11, 2020	296 hospitalized patients with COVID-19; 19 died	Diabetes Hypertension Smoking	Yes vs no Yes vs no Yes vs no	4.87 (1.70, 13.96) 1.82 (0.50, 6.63) 1.34 (0.16, 10.99)	Several variables not mentionned
Wang et al 2020, USA	Retrospective cohort study	Data from five member hospitals from the Mount Sinai Health System in New York City (NYC)	February 24 and April 15, 2020	3273 hospitalized patients with COVID-19; 742 died	BMI	per one unit increase	1.02 (1.00-1.03)	Age, sex and race
Wollenstein-Betech et al 2020, Brazil	Retrospective cohort study	Data from the nationwide surveilance program for acute respiratory distress syndrome (ARDS) in the state of São Paulo a	March 1 and June 30, 2020	113,214 hospitalized patients with COVID-19; 50,387 died	Diabetes Obesity	Yes vs. no Yes vs. no	1.22 (1.17-1.27) 1.66 (1.52-1.82)	None
Wu et al 2020, China	Retrospective cohort study	Data from Wuhan Hankou Hospital and No Six Hospital of Wuhan	December 26, 2019 and March 15 2020	1690 hospitalized patients with COVID-19; 556 died	Diabetes Hypertension Smoking	Yes vs. no Yes vs. no Yes vs. no	1.26 (0.72-2.22) 0.80 (0.52-1.22) 1.36 (0.89-2.08)	Age, gender, smoking, diabetes, hypertension, others comorbidites and severall biological measurements
Wu et al 2020, China	Retrospective cohort study	Data from to Wuhan Jinyintan Hospital	December 25, 2019 and January 26, 2020	201 hospitalized patients with COVID-19; 44 died	Diabetes Hypertension	Yes vs. no Yes vs. no	1.58 (0.80-3.13) 1.70 (0.92-3.14)	Age

Xie et al 2020, USA	Retrospective cohort study	Black patients from two hospitals in New Orleans	March 30 to 5 April 2020	287 hospitalized patients with COVID-19; 58 died	Diabetes Hypertension Obesity	Yes vs. no Yes vs. no Yes vs. no	1.92 (0.98-4.50) 2.71 (0.99-8.25) 1.67 (0.98-3.41)	Age, sex, race/ethnicity, hospital site, and the Charlson Comorbidity Index
Yahyavi et al 2020, Iran	Retrospective cohort study	Data from 17 referral hospitals affiliated to Iran University of medical sciences (IUMS)	February 20 and April 4, 2020	2553 hospitalized patients with COVID-19; 478 died	Diabetes	Yes vs. no	1.30 (1.00-1.60)	Angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, cardiovascular disease, chronic kidney disease, chronic pulmonary disease, diabetes mellitus, intensive care unit, diuretics, beta-blockers, and calcium channel blocker
Yan et al 2020, China	Retrospective cohort study	Patients from main tertiary teaching hospitals (Tongji Hospital, Optical Valley Branch and Sino-French New City Branch)	January 27 and February 17, 2020	882 hospitalized patients with COVID-19; 128 died	Diabetes	Yes vs. no	1.17 (0.73-1.86)	Age, sex, and comorbidities (cardiovascular disease, chronic respiratory disease, chronic kidney disease, cerebrovascular disease, diabetes, and malignancy), lymphopenia, d-dimer, lactate dehydrogenase, interleukin-6, myocardia injury, and liver injury
Yan et al 2020, China	Retrospective cohort study	Patients from Tongji Hospital (Wuhan)	January 10 to February 24, 2020	193 hospitalized patients with COVID-19; 108 died	Diabetes	Yes vs. no	1.53 (1.02-2.30)	Age, sex, hypertension, cardiovascular disease and cerebrovascular disease,
Yang et al 2020, China	Retrospective cohort study	Patients from the Central Hospital of Wuhan, a tertiary teaching hospital	January 1 to February 29, 2020	226 hospitalized patients with COVID-19; 50 died	Hypertension	Yes vs. no	2.68 (1.24-5.81)	Propensity score matching (multiple)
Yang et al 2020, China	Retrospective cohort study	Adult patients with COVID-19 hospitalized in Tongji Hospital (Wuhan)	February 4 to March 10, 2020	263 hospitalized patients with COVID-19; 107 died	Diabetes	Yes vs. no	1.59 (1.03-2.48)	Age

Yazdanpanah et al 2020, France	Cohort	French COVID cohort; patients from 25 hospitals participating in the cohort throughout France	January 24 and March 15, 2020	264 hospitalized patients with COVID-19; 42 died	Diabetes Obesity Hypertension Smoking	Yes vs. no Yes vs. no Yes vs. no Former vs. no Current vs. no	1.98 (1.01– 3.89) 3.32 (1.70-6.52) 3.77 (2.04 -6.99) 1.44 (0.65-3.22) 1.36 (0.31-5.86)	Age, sex, ethnicity, smoking and comorbidites
Yu et al 2020, China	Retrospective cohort study	Data from Tongji Hospital, Wuhan	January 14 and February 28, 2020	1663 hospitalized patients with COVID-19; 212 died	Diabetes Hypertension	Yes vs. no Yes vs. no	2.34 (1.45-3.76) 1.08 (0.68-1.72)	Age, sex, history of hypertension and diabetes, lymphopenia, increased alanine aminotransferase, increased lactate dehydrogenase, increased D-dimer, and increased PCT
Zandkarimi et al 2020, Iran	Cross-sectional Study	Kurdistan province COVID-19 patients, western Iran, (identified by 16-hour centers) admitted to hospitals (16 hospitals)	February 22 and May 18, 2020	1831 hospitalized patients with COVID-19; 142 died	Diabetes	Yes vs. no	2.16 (1.38-3.38)	Age, sex, CHD, CLD, CKD and cancer
Zangrillo et al 2020, Italy	Retrospective cohort study	Patients from the COVID-BioB study held in a 1350-bed university hospital in Milan and performed at Istituto di Ricovero e Cura a Carattere Scientifico (IRCCS) San Raffaele Scientific Institute	February 20 and April 2, 2020	73 hospitalized patients with COVID-19; 17 died	Hypertension	Yes vs. no	6.15 (1.75-29.11)	Age, sex and other variable not mentionned
Zhang et al 2020, China	Retrospective cohort study	multicenter hospitals in Wuhan (Department of Infectious Disease and the Department of Oncology of the Union Hospital of Tongji Medical College, Huazhong University of Science and Departments of Endocrinology in the following four hospitals; aged 38 to 66	Jan 1 to Mar 17, 2020	312 hospitalized patients with COVID-19; 33 died	Diabetes	Yes vs. no	6.87 (1.92-24.58)	Age, sex, hospital, aspartate aminotransferase, estimated glomerular filtration rate, prothrombin time, and procalcitonin
Zhang et al 2020, China	Retrospective cohort study	Data from Tongji Hospital and Wuhan Pulmonary Hospital	February 7 and March 27, 2020,	53 hospitalized patients with COVID-19; 13 died	BMI	BMI per 1 unit increase	1.35 (1.08-1.70)	None
Zhang et al	Retrospective	Patients admitted to Zhongnan	December 29,	289 hospitalized patients	Smoking	Yes vs. no	5.21 (1.39-19.52)	Age, chest

2020, China	cohort study	hospital of Wuhan Universityand No.7 hospital of Wuhan	2019 and February 16, 2020)	with COVID-19; 49 died				tightness/dyspnea, CRP and others biological parameters
Zhou et al 2020, China	Retrospective cohort study	Patients from Jinyintan Hospital and Wuhan Pulmonary Hospital (≥18 years old)		191 hospitalized patients with COVID-19; 54 died	Diabetes Hypertension Current smoker	Yes vs. no Yes vs. no Current smoker vs no	2.85 (1.35-6.05) 3.05 (1.57–5.92) 2.23 (0.65–7.63)	None
Zhu et al 2020, China	Retrospective cohort study	Data obtained from patient cohorts admitted in the 19 hospitals in Hubei Province, Chin	December 30, 2019 and March 20, 2020	7,337 hospitalized patients with COVID-19; 248 died	Diabetes	Yes vs. no	1.49 (1.13-1.96)	Age, gender, indicators of the severity of COVID-19 and hospital sites
Zimering et al 2020, USA	Retrospective cohort study	US military veterans hospitalized for COVID-19 infection during the peak of the pandemic in New Jersey	March 2020 and June 2020	55 hospitalized patients with COVID-19; 17 died	Diabetes BMI	Yes vs. no BMI per one unit increase	0.73 (0.23-2.35) 0.99 (0.92–1.07)	None

Supplementary Table 2: Risk of bias judgements for each studies from the ROBINS-I

Author, years, country	**Bias due to confounding**	**Bias due to selection of participants**	**Bias due to exposure assessment**	**Bias due to misclassification during follow-up**	**Bias due to missing data**	**Bias due to measurement of the outcome**	**Bias due to selective reporting of the results**	**Overall judgement**
Abohamr et al 2020, Saudi Arabia	Critical	Moderate	Low	Low	No information	Low	Moderate	Critical
Adrish et al, 2020, USA	Serious	Low	Low	Low	No information	Low	Low	Serious
Alamdari et al, 2020, Iran	Critical	Moderate	Moderate	Low	No information	Low	Low	Critical
Albitar et al, 2020, China	Low	Low	Low	Moderate	Low	Low	Moderate	Moderate
Alharthy et al, 2020, Saudi Arabia	Low	Moderate	Moderate	Low	Low	Low	Low	Moderate
Almazeedi et al 2020, Kuwait	No information	Low	Low	Low	No information	Low	Low	Low
Al-Salameh et al, 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
An et al 2020, South Korea	Low	Low	Low	Low	Low	Low	Low	Low
Anderson et al, 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Anzola et al, 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Arshad et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Atkins et al, 2020, UK	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Barbu et al, 2020, Romania	Serious	Low	Low	Low	No information	Low	Low	Serious
Baronio et al 2020, Italy	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Barron et al, 2020, England	Low	Low	Low	Low	Low	Low	Low	Low
Bepouka et al 2020, Democratic Republic of the Congo	Low	Low	Low	Low	Low	Low	Low	Low
Berenguer et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Biscarini et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Boulle et al, 2020, South Africa	Low	Low	Low	Low	No information	Low	Low	Low
Bousquet et al, 2020 France	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Brouns et al 2020, Netherlands	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Calmes et al 2020, Belgium	Moderate	Low	Low	Low	Moderate	Low	Low	Moderate
Cao et al, 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Carrasco-Sanchez et al, 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Castelnuovo et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Chachkhiani et al 2020, USA	Critical	Moderate	Low	Low	Moderate	Low	Low	Critical
Chang et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Chen et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Chen et al 2020, China	Critical	Low	Low	Low	Low	Low	Low	Critical
Chen et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low

Chilimuri et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Ciardullo et al 2020 Italy	Low	Low	Low	Low	Low	Low	Low	Low
Ciceri et al 2020, Spain	Serious	Low	Low	Low	Low	Low	Low	Serious
Coppelli et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Crouse et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Cummings et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Czernichow et al 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
De Souza et al 2020, Brazil	No information	Moderate	Low	Moderate	No information	Low	Low	Moderate
De Vito et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Deng et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Dennis et al 2020, England	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Desai et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Docherty et al 2020, UK	Low	Low	Low	Low	Low	Low	Low	Low
Du et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Escalera-Antezana et al 2020, Bolivia	Low	Moderate	Low	Moderate	No information	Low	Low	Moderate
Esme et al 2020, Turkey	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Fang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Farrel et al 2020, Ireland	Serious	Low	Low	Low	Low	Low	Low	Serious
Fernandez-Cruz et al 2020, Spain	Critical	Moderate	Low	Moderate	No information	Low	Low	Critical
Ferrando-Vivas et al 2020, Bolivia	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Foy et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Fried et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Galloway et al 2020, England	Moderate	Moderate	Moderate	Low	No information	Low	Low	Moderate
Gao et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Gayam et al 2020, USA	Low	Moderate	Moderate	Moderate	No information	Low	Moderate	Moderate
Gazzaruso et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Ghany et al 2020, USA	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Giacomelli et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Grasselli et al 2020, Italy	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Gu et al 2020, China	Serious	Low	Low	Low	Low	Low	Low	Serious
Gu et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Guisado-Vasco et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Gupta et al 2020, USA	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Gutierrez et al 2020, Mexico	Low	Low	Low	Low	Low	Low	Low	Low

Haase et al 2020, Denmark	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Hajifathalian et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Harmouch et al 2020, USA	Critical	Low	Low	Low	Low	Low	Low	Critical
Harrison et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Hashemi et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Hewitt et al 2020, EUROPE	Low	Low	Low	Low	Low	Low	Low	Low
Hu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Iaccarino et al 2020, Italy	No information	Low	Low	Low	No information	Low	Low	Low
Imam et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical
Ioannou et al 2020, USA	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Kabarriti et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Kim et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Kim et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Klang et al 2020, USA	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Klang et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Kolhe et al 2020, UK	Low	Low	Low	Low	Low	Low	Low	Low
Lampasona et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Lee et al 2020, South Korea	Critical	Moderate	Low	Low	No information	Low	Low	Critical
Li et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Li et al 2020, China	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Li et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Li et al 2020, China	No information	Low	Low	Low	No information	Low	Moderate	Moderate
Li et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Liu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Liu et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Liu et al 2020, China	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Liu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Lombardi et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Lopez-Mendez et al 2020, Mexico	Critical	Low	Low	Low	No information	Low	Low	Critical
Lunski et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Maddaloni et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Magleby et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Mallow et al 2020, USA	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Mather et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Matsuzawa et al 2020, Japan	Low	Low	Low	Low	No information	Low	Low	Low

McCarty et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
MCCULLOUGH et al 2020, USA	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Mendy et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Menezes-Soares et al 2020, Brazil	Critical	Low	Low	Low	No information	Low	Low	Critical
Meng et al 2020, China	No information	Low	Low	Low	Low	Low	Low	Low
Mikami et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Mirani et al 2020, Italy	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Moon et al 2020, South Korea	Low	Low	Low	Low	Low	Low	Low	Low
Moon et al 2020, South Korea	Serious	Low	Low	Low	No information	Low	Low	Serious
MOWLA et al 2020, Bangladesh	Low	Low	Low	Moderate	No information	Low	Low	Moderate
Munblit et al 2020, Russia	Low	Low	Low	Low	No information	Low	Low	Low
Munoz et al 2020, Spain	Low	Low	Low	Low	No information	Low	Low	Low
Munoz-price et al 2020, USA	Low	Low	Low	Moderate	Moderate	Low	Low	Moderate
Nachega et al 2020, Congo	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Nakeshbandi et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Namendys-Silva et al 2020, Mexico	Critical	Moderate	Low	Low	Moderate	Low	Low	Critical
Narain et al 2020, USA	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Neumann-Podczaska et al 2020, Poland	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Nicholsoni et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Nie et al 2020, China	Critical	Low	Low	Low	No information	Low	Low	Critical
Nogueira et al 2020, Portugal	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Núñez-Gil et al 2020, Spain	Critical	Low	Low	Low	No information	Low	Moderate	Critical
Palaiodimos et al 2020, USA	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Pan et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Park et al 2020, South Korea	Low	Low	Low	Low	No information	Low	Low	Low
Parker et al 2020, South Africa	Critical	Low	Low	Low	No information	Low	Moderate	Critical
Perkin et al 2020, England	Serious	Low	Low	Low	No information	Low	Low	Serious
Petrilli et al 2020, USA	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Pettit et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Poblador-Plou et al 2020, Spain	Serious	Low	Low	Low	No information	Low	Low	Serious
Portoles et al 2020, Spain	Serious	Low	Low	Low	No information	Low	Low	Serious
Price-Haywood et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Qin et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Ramos-Rincon et al 2020, Spain	Critical	Moderate	Low	Low	No information	Low	Low	Critical
Rapp et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low

Rastad et al 2020, Iran	Low	Low	Low	Low	Moderate	Low	Low	Low
Recinella et al 2020, Italy	Serious	Moderate	Low	Low	Low	Low	Low	Serious
Reilev et al 2020, Denmark	Moderate	Low	Low	Low	No information	Low	Low	Moderate
Rivera-Izquierdo et al 2020, Spain	Low	Low	Low	Low	No information	Low	Low	Low
Rodrıguez-Molinero et al 2020, Spain	Critical	Low	Low	Low	Low	Low	Low	Critical
Rosenberg et al 2020, USA	Serious	Low	Low	Low	No information	Low	Low	Serious
Rossi et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Rottoli et al 2020, Italy	Low	Low	Low	Low	Low	Low	Low	Low
Rozaliyani et al 2020, Indonesia	Low	Low	Low	Low	No information	Low	Low	Low
Russo et al 2020, Italy	Low	Low	Low	Low	No information	Low	Low	Low
Salacup et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Salazar et al 2020, USA	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Sands et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Santos et al 2020, Brazil	Low	Low	Low	Low	Low	Low	Low	Low
Seiglie et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Shah et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Shang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Shi et al 2020, China	Moderate	Low	Low	Low	Moderate	Low	Low	Moderate
Shi et al 2020, China	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Smith et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Sousa et al 2020,Brazil	No information	Low	Low	Low	Moderate	Low	Low	Moderate
Steinberg et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical
Stony Brook COVID-19 Research Consortium et al 2020, USA	Critical	Low	Moderate	Low	Low	Low	Low	Critical
Sun et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Talavera et al 2020, Spain	Low	Low	Low	Low	Low	Low	Low	Low
Tartof et al 2020, USA	Low	Low	Low	Low	Low	Low	Low	Low
Tehrani et al 2020, Sweden	Critical	Low	Low	Low	No information	Low	Low	Critical
Turcotte et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Wang et al 2020, China	Serious	Moderate	Low	Low	No information	Low	Low	Serious
Wang et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Wang et al 2020, China	Low	Moderate	Low	Low	Moderate	Low	Low	Moderate
Wang et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Wang et al 2020, USA	Low	Low	Low	Low	No information	Low	Low	Low
Wollenstein-Betech et al 2020, Brazil	Critical	Low	Low	Low	Moderate	Low	Low	Critical

Wu et al 2020, China	Low	Moderate	Low	Low	Low	Moderate	Low	Moderate
Wu et al 2020, China	Serious	Low	Low	Low	No information	Low	Low	Serious
Xie et al 2020, USA	Low	Low	Low	Low	No information	Low	Moderate	Moderate
Yahyavi et al 2020, Iran	Low	Low	Low	Low	No information	Low	Low	Low
Yan et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Yan et al 2020, China	Low	Moderate	Low	Low	No information	Low	Low	Moderate
Yang et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Yang et al 2020, China	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Yazdanpanah et al 2020, France	Low	Low	Low	Low	Low	Low	Low	Low
Yu et al 2020, China	Low	Low	Low	Low	No information	Low	Low	Low
Zandkarimi et al 2020, Iran	No information	Low	Low	Moderate	No information	Low	Low	Moderate
Zangrillo et al 2020, Italy	No information	Moderate	Low	Low	No information	Low	Moderate	Moderate
Zhang et al 2020, China	Low	Low	Low	Low	Low	Low	Low	Low
Zhang et al 2020, China	Critical	Moderate	Moderate	Low	No information	Low	Moderate	Critical
Zhang et al 2020, China	No information	Moderate	Low	Low	Moderate	Low	Low	Moderate
Zhou et al 2020, China	Serious	Low	Low	Low	No information	Low	Low	Serious
Zhu et al 2020, China	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Zimering et al 2020, USA	Critical	Low	Low	Low	No information	Low	Low	Critical

Supplementary Table 3. Subgroup analyses of association between BMI and mortality risk in patients with COVID-19										
	Obesity (obese vs non-obese)					**BMI per 5 kg/m^2**				
	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]
All studies	54	1.45 (1.31-1.61)	91.0	<0.0001		25	1.12 (1.07-1.17)	68.2	<0.0001	
Study design										
Retrospective	47	1.44 (1.29-1.61)	92.1	<0.0001		20	1.11 (1.06-1.16)	59.8	<0.0001	
Prospective	4	1.45 (1.07-1.97)	34.1	0.208		3	1.15 (1.02-1.30)	84.8	0.001	
Cross-sectional	3	1.88 (0.87-4.02)	61.8	0.07	0.85	2	1.50 (0.68-3.31)	83.5		0.89
Geographical location										
North America	26	1.26 (1.09-1.46)	94.2	<0.0001		16	1.13 (1.07-1.19)	52.3	<0.0001	
South America	4	1.80 (1.24-2.62)	23.7	0.27		1	1.09 (1.02-1.16)			
Europe	17	1.40 (1.23-1.60)	46.2	0.02		7	1.14 (1.01-1.29)	84.2	<0.0001	
Asia	5	2.50 (0.72-8.68)	95.5	<0.0001		1	0.99 (0.83-1.17)			
Africa	2	1.93 (1.15-3.23)	0.5	0.32	0.07					0.81
Number of patients										
<1000	31	1.76 (1.42-2.18)	87.2	<0.0001	0.0001	13	1.19 (1.07-1.31)	62.6	0.001	
≥1000	23	1.27 (1.13-1.42)	88.3	<0.0001		11	1.10 (1.05-1.15)	70.1	<0.0001	0.001
Risk of bias										
Low	28	1.39 (1.20-1.60)	93.3	<0.0001		12	1.12 (1.06-1.18)	70.3	<0.0001	
Moderate	11	1.52 (1.19-1.95)	68.9	<0.0001		12	1.15 (1.06-1.25)	66.7	0.001	
Serious	3	0.97 (0.78-1.21)	0.0	0.675		1	0.76 (0.49-1.20)			0.36
Critical	12	1.75 (1.32-2.32)	91.9	<0.0001	0.42					
Adjustment for confounders										
No	12	1.69 (1.30-2.21)	91.3	<0.0001		1	1.05 (0.88-1.25)	-		
Age only	3	0.97 (0.78-1.21)	0.0	0.675	0.31	1	0.76 (0.49-1.20)	-		
Multiple	36	1.43 (1.26-1.61)	91.6	<0.0001		23	1.13 (1.08-1.18)	69.8	<0.0001	0.35

Footnote: CI confidence interval; SRR summary relative risk

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-values of 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity within each subgroup

[b]P-value for heterogeneity between subgroups with meta-regression analysisbetween

Supplementary Table 4. Subgroup analyses of association between smoking status and mortality risk in patients with COVID-19

	Smoking (ever versus never smoker)					Current smoker (yes vs no)					Ex-smoker (yes vs no)				
	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]	n	SRR (95% CI)	I^2 (%)	P_{within} [a]	$P_{between}$ [b]
All studies	28	1.28 (1.17-1.40)	74.0	<0.0001		19	1.29 (1.03-1.62)	84.2	<0.0001		14	1.25 (1.11-1.42)	74.8	<0.0001	
Study design															
Retrospective	21	1.32 (1.15-1.52)	75.9	<0.0001		14	1.41 (1.04-1.91)	86.2	<0.0001		10	1.33 (1.12-1.57)	74.6	<0.0001	
Prospective	5	1.16 (1.01-1.32)	65.8	0.020		5	1.07 (0.94-1.21)	0.0	0.83		4	1.15 (0.93-1.42)	76.8	0.005	
Cross-sectional	2	2.65 (0.52-13.38)	81.7	0.020	0.45	-	-	-		0.01		-	-	-	
Geographical location															0.007
North America	17	1.25 (1.08-1.45)	79.4	<0.0001		11	1.16 (0.99-1.35)	34.3	0.125		9	1.23 (1.06-1.43)	71.9	<0.0001	
Europe	9	1.26 (1.13-1.40)	59.2	0.012		5	1.07 (0.95-1.21)	0.0	0.446		4	1.30 (1.00-1.69)	76.3	<0.0001	
Asia	2	2.56 (0.99-6.59)	53.0	0.15		3	3.00 (2.50-3.61)	0.0	0.82		1	2.06 (0.73-5.79)			0.71
South America	_	_	_	_		-	-	-				-	-	-	
Africa	_	_	_	_	0.21	-	-	-		<0.0001		-	-	-	
Number of patients															
<1000	13	1.47 (1.19-1.80)	46.1	0.035		6	2.47 (1.68-3.62)	27.1	0.23	<0.0001	4	1.91 (1.48-2.48)	0.0	0.784	
≥1000	15	1.22 (1.10-1.37)	82.1	<0.0001	0.001	13	1.11 (1.00-1.24)	27.1	0.17		10	1.17 (1.04-1.33)	76.9	<0.0001	0.002
Risk of bias															
Low	12	1.20 (1.08-1.34)	46.4	0.039		8	1.09 (0.98-1.20)	0.0	0.854		6	1.26 (1.07-1.49)	71.5	0.004	
Moderate	10	1.25 (1.04-1.51)	87.3	<0.0001		6	1.23 (0.69-2.22)	93.3	<0.0001		4	1.12 (0.90-1.39)	75.8	0.006	
Serious	3	1.43 (1.24-1.66)	0.0	0.495		3	1.57 (1.23-2.01)	0.0	0.674		3	1.32 (1.07-1.64)	10.4	0.327	
Critical	3	1.51 (0.99-2.30)	40.3	0.19	0.60	2	2.50 (1.37-4.58)	1.6%	0.313	0.18	1	1.82 (1.27-2.60)			0.53
Adjustment for confounders															
No	4	1.43 (1.13-1.80)	44.4	0.15		3	2.45 (1.43-4.19)	0.0	0.594		1	1.82 (1.27-2.60)			
Age only	3	1.43 (1.24-1.66)	0.0	0.495	0.40	2	1.55 (1.21-1.99)	0.0	0.493		3	1.32 (1.07-1.64)	10.4	0.327	
Multiple	21	1.23 (1.09-1.38)	78.3	<0.0001		14	1.16 (0.88-1.52)	87.6	<0.0001	0.13	10	1.20 (1.05-1.39)	79.6	<0.0001	0.41

Footnote: CI confidence interval; SRR summary relative risk

I^2 (%) is a measure of the proportion of the heterogeneity attributed to between study variation rather than due to chance. I^2-values of 25%, 50% and 75% indicates low, moderate and high between study heterogeneity, respectively.

[a]P-value for heterogeneity within each subgroup

[b]P-value for heterogeneity between subgroups with meta-regression analysisbetween

Supplementary Table 5: GRADE Evidence profile

		Downgrading factors					Upgrading factors					
Prognostic factors	Initial rate of confidence	Risk of bias	Inconsistency	Indirectness	Imprecision (ratio of the upper to lower 95%CI)	Publication bias	Large magnitude of association	Dose response	Residual confounding	Consistency	Overall quality	Level of evidence for the health effect
Diabetes	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio: 0.97)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⨁⨁◯◯ Low	Low
Hypertension	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.18)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⨁⨁◯◯ Low	Low
Obesity	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.23)	Not likely	Upgrade (SRR>2 for BMI ~ 45)	Upgrade (evidence of nonlinear relation)	No upgrade	No upgrade	⨁⨁⨁⨁ High	High
Smoking	Low (observational studies)	Not likely	Not likely	Not likely	Not likely (ratio : 1.20)	Not likely	No upgrade	No upgrade	No upgrade	No upgrade	⨁⨁◯◯ Low	Low

Supplementary method for GRADE

GRADE (Grading of Recommendations, Assessment, Development and Evaluations) is a transparent framework for developing and presenting summaries of evidence and provides a systematic approach for making clinical practice recommendations. The aim is to rate the certainty of evidence (or known as quality of evidence) for each outcome across studies (i.e. a body of evidence) using four levels of evidence: very low, low, moderate, and high. Guidelines are described in a serie of articles published in the Journal of Clinical Epidemiology: https://www.jclinepi.com/action/showPdf?pii=S0895-4356%2810%2900330-6

GRADE does not mean rating each study as a single unit. Rather, rating is made for each outcome, and quality may differ indeed, is likely to differ from one outcome to another within a single study and across a body of evidence.

Quality of evidence	Interpretation
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect
Very Low	We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect

In the GRADE framework, observational study initially starts as low-quality evidence. Then, there are 5 criteria which can be used to downgrade the level of evidence:

- Risk of bias in individual studies – e.g. methodological issues in included studies such as inadequate blinding (participants knew they were in control/treatment group)
- Inconsistency of results between studies
- Indirectness of evidence – e.g. participants were children although the systematic review was about adults
- Imprecision – results were not statistically significant, or the effect was clinically important once the studies were meta-analysed
- Publication bias – result was biased due to a file-drawering effect, as studies not showing a statistically significant effect are less likely to be published.

Additionally, observational studies starting at low can be upgraded based on 3 criteria: large effect, dose-response effect and “Effect of all plausible confounding factors would be to reduce the effect (where an effect is observed) or suggest a spurious effect (when no effect is observed)”.

Guidance to rate the outcome is briefly described below:

1- Risk of bias across studies (internal validity)

Risk of bias was assessed using ROBIN-I rating the overall risk of bias into four categories: low, moderate, serious and critical risk of bias.

Downgrade	Guidance
Not likely	Most information is at low risk of bias
Serious	Does not clearly meet guidance for “not serious” or “very serious”
Very serious	Most information is at serious or critical risk of bias

2- Inconsistency

Magnitude of intervention effects differs across studies, it may be explained by the difference in population, intervention, outcomes or study methods. If large variability (heterogeneity) in magnitude of effect remains unexplained, the quality of evidence decreases.

Downgrade	Guidance
Not likely	Point estimates similar Confidence intervals overlap Statistical heterogeneity is non-significant ($p \geq 0.1$) I^2 of $\leq 50\%$
Serious	Point estimates vary Confidence intervals show minimal overlap Statistical heterogeneity has low p-value ($p \leq 0.1$) I^2 of >50% to 75%
Very serious	Point estimates vary widely Confidence intervals show minimal or no overlap Statistical heterogeneity has low p-value ($p \leq 0.1$) I^2 of >75%

3- Directness

Certainty can be rated down if the patients studied are different from those for whom the recommendation applies. Decisions regarding indirectness of patients and interventions depend on an understanding of whether biological or social factors are sufficiently different that one might expect substantial differences in the magnitude of effect. Indirectness occurs when the population, intervention, or outcomes differ from those in which we are interested

4- Precision is the degree of certainty surrounding an effect estimate with respect to a given outcome (AHRQ 2012a). A precise estimate enables the evaluator to determine whether there is an effect (i.e., it is different from the comparison group). We used 95% confidence intervals as the primary method to assess imprecision (Guyatt et al. 2011b).

Downgrade	Guidance
Not likely	No or minimal indications of large standard deviations (i.e., SD > mean) For ratio measures (e.g., odds ratio, OR) the ratio of the upper to lower 95% CI for most studies (or meta-estimate) is < 10; or for absolute measures (e.g., percent control response) the absolute difference between the upper and lower 95% CI for most studies (or meta-estimate) is < 100
Serious	Does not clearly meet guidance for "not serious" or "very serious"
Very serious	Large standard deviations (i.e., SD > mean) For ratio measures (e.g., OR) the ratio of the upper to lower 95% CI for most studies (or meta-estimate) is ≥ 10; or for absolute measures (e.g., percent control response) the absolute difference between the upper and lower 95% CI for most studies (or meta-estimate) is ≥ 100

5- Publication bias

We characterized publication bias as "undetected" (no downgrade) or "strongly suspected" as recommended by GRADE (Guyatt et al. 2011e). In general, studies with statistically significant results are more likely to be published than studies without statistically significant results (''negative studies'') (Guyatt et al. 2011e). Thus, some degree of publication bias is likely on any topic; however, downgrading is reserved for cases where the concern is serious enough to significantly reduce confidence in the body of evidence.

Publication bias is suspected when studies are uniformly small, particularly when sponsored by industries, NGOs or authors with conflicts of interests. Funnel plots and Egge's regression can help to visualize publication bias.

6- Domains that can increase confidence

Upgrade	Guidance
Large magnitude of association or effect	GRADE Working Group has previously suggested guidelines for rating quality of evidence up by one category (typically from low to moderate) for associations greater than 2, and up by two categories for associations greater than 5 (Guyatt et al.2011g).
Dose response	Upgrade for evidence of a monotonic dose-response gradient (Guyatt et al. 2011g) and for evidence of a non-monotonic dose response when data fit the expected pattern, i.e., prior knowledge leads to expectation for non-monotonic dose response, and/or non-monotonic dose response is consistently observed in the evidence base.
Residual Confounding or Other Related Factors	Confidence in the results is increased when the body of evidence is potentially biased by factors counter to the observed effect. Upgrading should be considered when there are indications that residual confounding or bias would underestimate an apparent association or treatment effect (i.e., bias towards the null), or suggest a spurious effect when results suggest no effect.
Cross-Species/Population/Study Consistency	Three types of consistency in the body of evidence can be used to support an increase in confidence in the results: •across animal studies–consistent results reported in multiple experimental animal models or species •across dissimilar populations–consistent results reported across populations (human or wildlife) that differ in factors such as time, location, and/or exposure •across study types–consistent results reported from studies with different design features, e.g., between prospective cohort and case-control human studies or between chronic and multigenerational animal studies

Translation of confidence rating into level of evidence		
Confidence in the body of evidence	Direction of the effect (Health effect)	Level of evidence for the health effect
High	⇨	High
Moderate	⇨	Moderate
Low	⇨	Low
Very low or no evidence	⇨	Inadequate
Confidence in the body of evidence	Direction of the effect (No Health effect)	Level of evidence for the health effect
High	⇨	Evidence of no health effect
Moderate	⇨	Inadequate
Low	⇨	Inadequate
Very low or no evidence	⇨	Inadequate